MIL-M-28787/141B(EC)
23 August 1983
SUPERSEDING
MIL-M-28787/141A(EC)
2 June 1977

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

MEMORY, RANDOM ACCESS, DIGITAL, KEY CODE GYB

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic random access memory module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

* 2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.727 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be GYB.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/141-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 V dc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 V dc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

* 4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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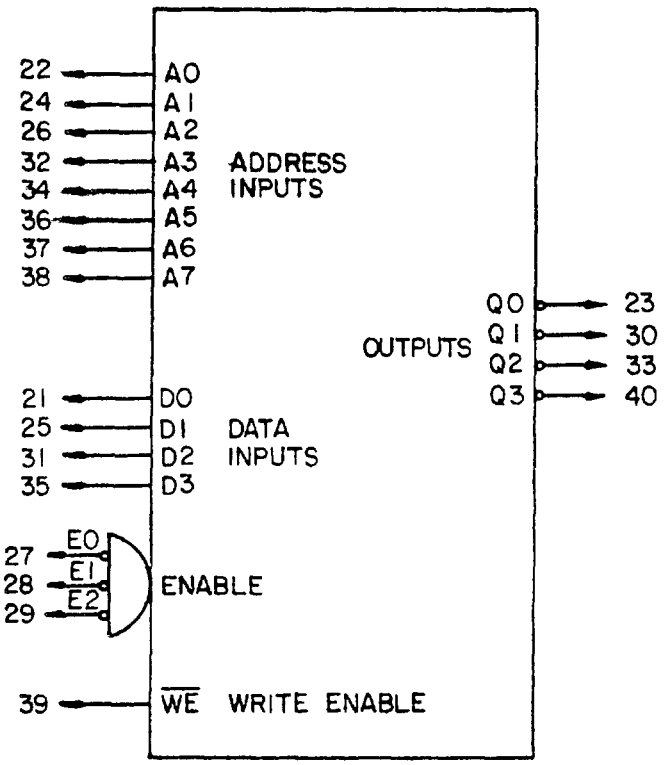


*FIGURE 1. GYB module function diagram, random access memory.

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*TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A4	V_{OUT} (low loaded)	II	S	0	900	0	900	0	900	0	900	mV
B1-B8, B13-B16	Input breakdown current	III	S	0	4	0	4	0	4	0	4	mA
B9-B12	Input breakdown current	III	S	0	1	0	1	0	1	0	1	mA
C1-C16	Input clamp diode voltage	IV	X	-0.2	-1.5	-0.2	-1.5	-0.2	-1.5	-0.2	-1.5	V
D1	Isolation, frame	V	S	10	∞	1	∞	10	∞	1	∞	MΩ
D2-D20	Isolation, imused pins	V	X	10	∞	1	∞	10	∞	1	∞	MΩ
E1-E4	Output voltage, low level	VI	X	0	450	0	450	0	450	0	450	mV
F1-F4	Output voltage, high level	VII	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
G1-G4	Address threshold, low level	VIII	X	0	450	0	450	0	450	0	450	mV
G5-G8	Address threshold, high level	VIII	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
H1-H8, H13-H16	Input current, low level	IX	X	0.0	3.5	0.0	3.5	0.0	3.5	0.0	3.5	mA
H9-H12	Input current, low level	IX	X	0.00	0.87	0.00	0.87	0.00	0.87	0.00	0.87	mA
J1-J8, J13-J16	Input current, high level	X	X	-1,600	100	-1,600	100	-1,600	100	-1,600	100	µA
J9-J12	Input current, high level	X	X	-400	25	-400	25	-400	25	-400	25	µA
K1-K4	Output short circuit current	XI	X	20	100	20	100	20	100	20	100	mA
K5-K8	Output leakage current	XI	X	-100	100	-100	100	-100	100	-100	100	µA
K9-K12	Output leakage current	XI	X	0	100	0	100	0	100	0	100	µA
L1-L3	Power supply current	XII	X	5	455	5	455	5	455	5	455	mA

See footnotes at end of table.

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*TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
M1-M4	Transition time (t_{TLH}), C_L = 50 pF	XIII	S	5	25	5	25	5	25	5	25	ns
M5-M8	Transition time (t_{TLH}), C_L = 150 pF	XIII	Q	5	35	5	35	5	35	5	35	ns
N1-N4	Transition time (t_{THL}), C_L = 50 pF	XIII	S	2	15	2	15	2	15	2	15	ns
N5-N8	Transition time (t_{THL}), C_L = 150 pF	XIII	Q	3	25	3	25	3	25	3	25	ns
P1-P4	Enable recovery time (t_{ERL}), low-state to high-impedance state	XIV	S	1	40	1	40	1	40	1	40	ns
Q1-Q4	Enable recovery time (t_{ERH}), high-state to high-impedance state	XIV	S	1	40	1	40	1	40	1	40	ns
R1	Walking-one-walking-zero test pattern (C_L = 50 pF, V_{CC} = 4.5 V)	XV	X	← See table XV →								
R2	Walking-one-walking-zero test pattern (C_L = 50 pF, V_{CC} = 5.5 V)	XV	X	← See table XV →								
R3	Walking-one-walking-zero test pattern (C_L = 150 pF, V_{CC} = 4.5 V)	XV	Q	← See table XV →								
R4	Walking-one-walking-zero test pattern (C_L = 150 pF, V_{CC} = 5.5 V)	XV	Q	← See table XV →								
S1	Galloping-one-galloping-zero test pattern (C_L = 50 pF, V_{CC} = 4.5 V)	XVI	X	← See table XVI →								

See footnotes at end of table.

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*TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
S2	Galloping-one-galloping-zero test pattern (C_L = 50 pF, V_{CC} = 5.5 V)	XVI	X	← See table XVI →								
S3	Galloping-one-galloping-zero test pattern (C_L = 150 pF, V_{CC} = 4.5 V)	XVI	Q	← See table XVI →								
S4	Galloping-one-galloping-zero test pattern (C_L = 150 pF, V_{CC} = 5.5 V)	XVI	Q	← See table XVI →								
T1	Inter-address write-recovery-one-write-recovery-zero test pattern (C_L = 50 pF, V_{CC} = 4.5 V)	XVII	X	← See table XVII →								
T2	Inter-address write-recovery-one-write-recovery-zero test pattern (C_L = 50 pF, V_{CC} = 5.5 V)	XVII	X	← See table XVII →								
T3	Inter-address write-recovery-one-write-recovery-zero test pattern (C_L = 150 pF, V_{CC} = 4.5 V)	XVII	Q	← See table XVII →								
T4	Inter-address write-recovery-one-write-recovery-zero test pattern (C_L = 150 pF, V_{CC} = 5.5 V)	XVII	Q	← See table XVII →								
U1	Marching write-recovery-one-write-recovery-zero and enable/disable test pattern (C_L = 50 pF, V_{CC} = 4.5 V)	XVIII	X	← See table XVIII →								
U2	Marching write-recovery-one-write-recovery-zero and enable/disable test pattern (C_L = 50 pF, V_{CC} = 5.5 V)	XVIII	X	← See table XVIII →								

See footnotes at end of table

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*TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
U3	Marching write-recovery-one-write-recovery-zero and enable/disable test pattern (C_L = 150 pF, V_{CC} = 4.5 V)	XVIII	Q	← See table XVIII →								
U4	Marching write-recovery-one-write-recovery-zero and enable/disable test pattern (C_L = 150 pF, V_{CC} = 5.5 V)	XVIII	Q	← See table XVIII →								
V	Power supply transient amplitude	XIX	Q	0	400	0	400	0	400	0	400	mV
W1-W3	Input crosstalk, low level	XX	Q	0	700	0	700	0	700	0	700	mV
W4	Input crosstalk, low level	XX	Q	0	600	0	600	0	600	0	600	mV
W5-W7	Input crosstalk, high level	XX	Q	0	600	0	600	0	600	0	600	mV
W8	Input crosstalk, high level	XX	Q	0	500	0	500	0	500	0	500	mV
X	Power supply filter capacitance	XXI	S	1.65	2.75	1.65	2.75	1.65	2.75	1.65	2.75	µF
Y1-Y8, Y13-Y15	Input capacitance	XXII	Q	2	45	2	45	2	45	2	45	pF
Y9-Y12	Input capacitance	XXII	Q	2	17	2	17	2	17	2	17	pF
Y16	Input capacitance	XXII	Q	2	50	2	50	2	50	2	50	pF
Y17-Y20	Output capacitance (Q)	XXII	Q	2	18	2	18	2	18	2	18	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.

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4.4.2 Order of testing. Test conditions A through Y shall be performed in any order with the following exceptions. If test conditions A, B, C, or D are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D where applicable, followed by the remaining tests in the sequence.

4.4.3 Functional test procedures. To ensure that device characteristics, as specified in 6.4, are guaranteed, the tester's nominal setting is offset to allow for tester inaccuracies using LSI tester. The inaccuracies are comprised of the initial calibration tolerances and the inherent dynamic skew of the drive circuitry. The calibration inaccuracy consists of the timing-edge to timing-edge adjustment tolerance. The dynamic inaccuracies are comprised of the inherent interaction between timing and driver circuits. To derive the nominal setting of the timing edges, the total inaccuracy (initial calibration and dynamic skew) is subtracted from the guaranteed minimum or maximum timing. The inaccuracy figures will be approved by the Standard Electronic Modules Program Quality Assurance Activity (SEMP-QAA) as part of the module correlation procedures.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through Y in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions C and E through X shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 5.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 6. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3. Following the life test and while still at the maximum class temperature, electrical tests E through Y shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests E through Y shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Cross reference of classification. Part number M28787/141-1 supersedes part number M28787/141. The two parts are interchangeable.

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6.4 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life limits from table I.

Maximum power dissipation	2.5 W (outputs not loaded)
Supply voltage	5 V dc ±10 percent
Input voltage	
Low level	-0.5 V minimum, 0.8 V maximum
High level	2 V minimum, V_{CC} maximum
Input current (D0 - D3)	
Low level (V_{IL} = 0 V)	0.87 mA maximum
(V_{IL} = 0.4 V)	0.8 mA maximum
High level (V_{IH} = 2.4 V)	25 µA maximum
(V_{IH} = 5.5 V)	1 mA maximum
Input current (A0 - A7, $\overline{WE}$, E0 - E2)	
Low level (V_{IL} = 0 V)	3.5 mA maximum
(V_{IL} = 0.4 V)	3.2 mA maximum
High level (V_{IH} = 2.4 V)	100 µA maximum
(V_{IH} = 5.5 V)	4 mA maximum
Input clamp diode voltage	
V_{CC} = 4.5 V, I_{CLAMP} = -7.5 mA	-0.2 V minimum, -1.5 V maximum
Output voltage (Q0 - Q3)	
Low level (I_{OL} = 10 mA)	0.45 V maximum
High level (I_{OH} = -3.2 mA)	2.4 V minimum
Output drive current (Q0 - Q3)	
Maximum capability	
Low level (V_{OL} = 0.45 V)	10 mA
High level (V_{OH} = 2.4 V)	-3.2 mA
Output leakage current	
High-impedance output state	
V_{OL} = 0.45 V	-100 µA minimum, 100 µA maximum
V_{OH} = 5.5 V	100 µA maximum
Input capacitance	
D0 - D3, data	15 pF maximum
A0 - A7, address	45 pF maximum
E0 - E3, enable	45 pF maximum
$\overline{WE}$, write enable	45 pF maximum
Output capacitance (Q0 - Q3)	18 pF maximum
Dynamic characteristics (reference figure 2)	
Setup time	
Address-to-write enable (t_{AS})	15 ns minimum
Enable-to-write enable (t_{ES})	15 ns minimum
Data-to-write enable (t_{DS})	15 ns minimum
Hold time	
Address-to-write enable (t_{AH})	10 ns minimum
Enable-to-write enable (t_{EH})	10 ns minimum
Data-to-write enable (t_{DH})	10 ns minimum
Write pulse width (t_{WP})	65 ns minimum
Address access time (t_{AA})	
Address-to-output	
C_L = 50 pF load	70 ns maximum
C_L = 150 pF load	80 ns maximum

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Dynamic characteristics (reference figure 2) - Continued	
Enable access time (t_{EA})	
Enable-to-output	
C_L = 50 pF load	45 ns maximum
C_L = 150 pF load	55 ns maximum
Enable recovery time ($C_L \leq$ 10 pF load)	
Low-state-to-high-impedance state (t_{ERL})	40 ns maximum
High-state-to-high-impedance state (t_{ERH})	40 ns maximum
Write-recovery-propagation-delay time (t_{WR})	
C_L = 50 pF load	120 ns maximum
C_L = 150 pF load	130 nx maximum

* 6.5 Module function. This module contains a random access memory of 256 words by 4 bits with three-state outputs.

6.5.1 Input to output relationship. The following describes the essential random access memory (RAM) operation. Figure 2 illustrates the input to output relationship.

6.5.1.1 Read. The memory is addressed with A0 - A7 inputs which selects one of 256 words. The chip is enabled by making all enables (E0, E1, E2) low. If the write enable ($\overline{WE}$) is high and the chip is enabled the stored data is read out on the data out pins. The data readout is the complement of the data written in during the write cycle.

6.5.2.1 Write. The memory is addressed with A0 - A7 inputs which selects one of the 256 words. The chip is enabled as in the read cycle. If the write enable is low the data on the data input pins is written into the addressed word. The output state is undefined in the write mode when $\overline{WE}$ is low, the output could be active or in the high-impedance state.

6.5.1.3 Three-state outputs. The three-state output mode allows the module to be connected to a common bus with several like modules. The memory can be placed in the high-impedance state by setting any of the enables at a high level.

6.5.2 Parallel outputs. Bussing is allowed between like outputs of a number of modules. When outputs of several GYB modules are bussed together, all address lines should be addressed using synchronous clocking. Outputs of a single module cannot be paralleled.

6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

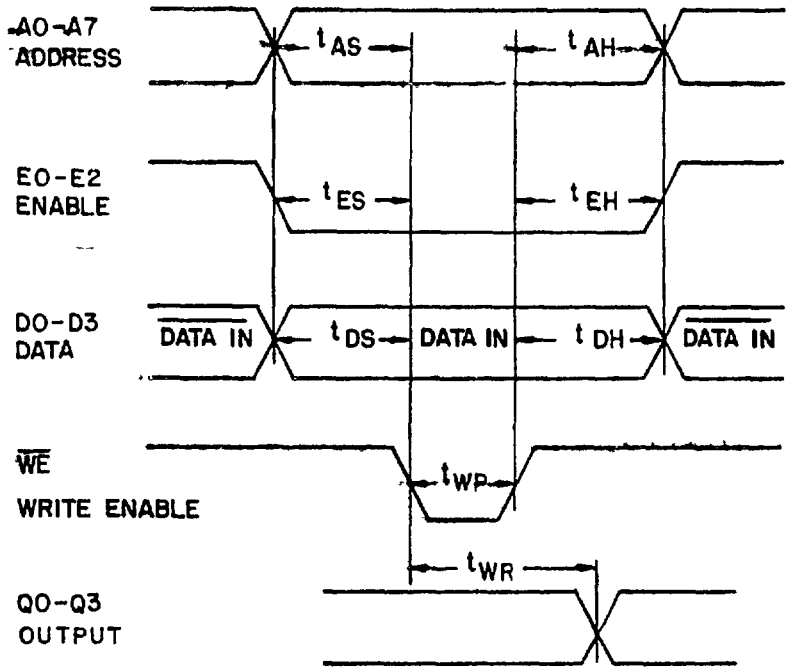
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Preparing activity:
Navy - EC

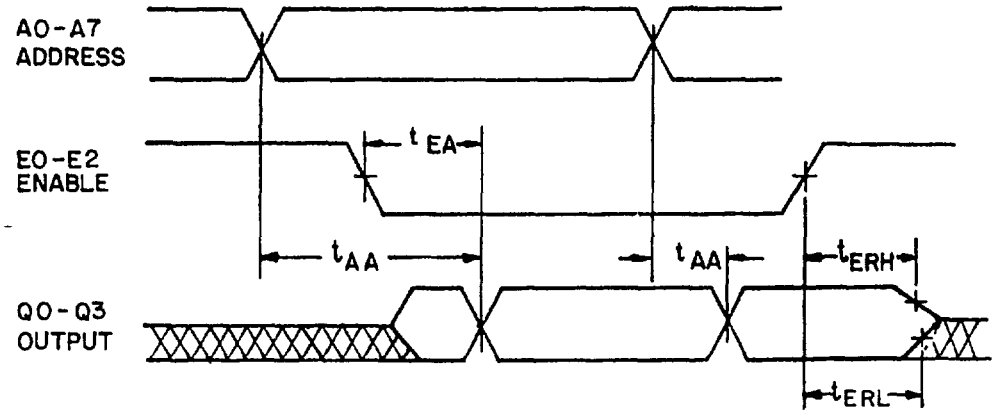
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WRITE CYCLE



READ CYCLE

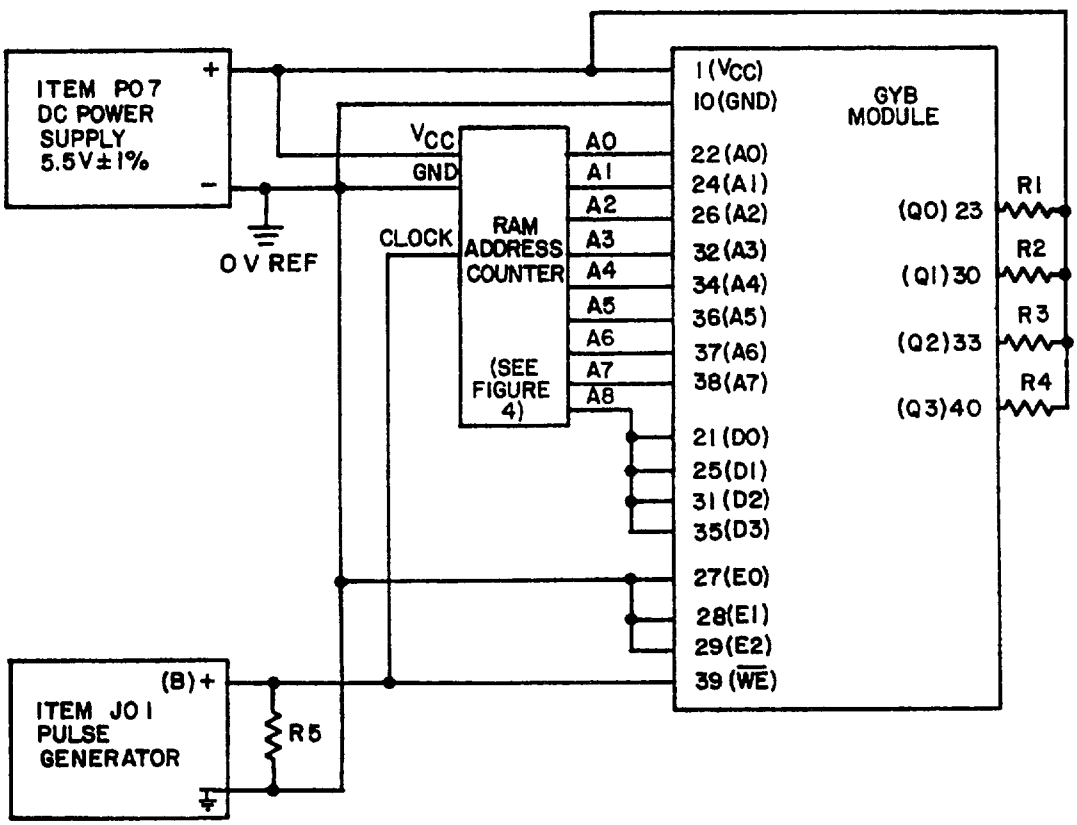


Inputs					Output	Mode
E0	E1	E2	$\overline{WE}$	D_{IN}	Q	
H	X	X	X	X	High Z	Not selected
X	H	X	X	X	High Z	Not selected
X	X	H	X	X	High Z	Not selected
L	L	L	L	L	Undefined	Write L
L	L	L	L	H	Undefined	Write H
L	L	L	H	X	$\overline{D_{IN}}$	Read inverted data from addressed location

H = High level
L = Low level
X = Don't care (high or low)
High Z = High-impedance state

*FIGURE 2. Dynamic characteristics waveforms and GYB truth table.

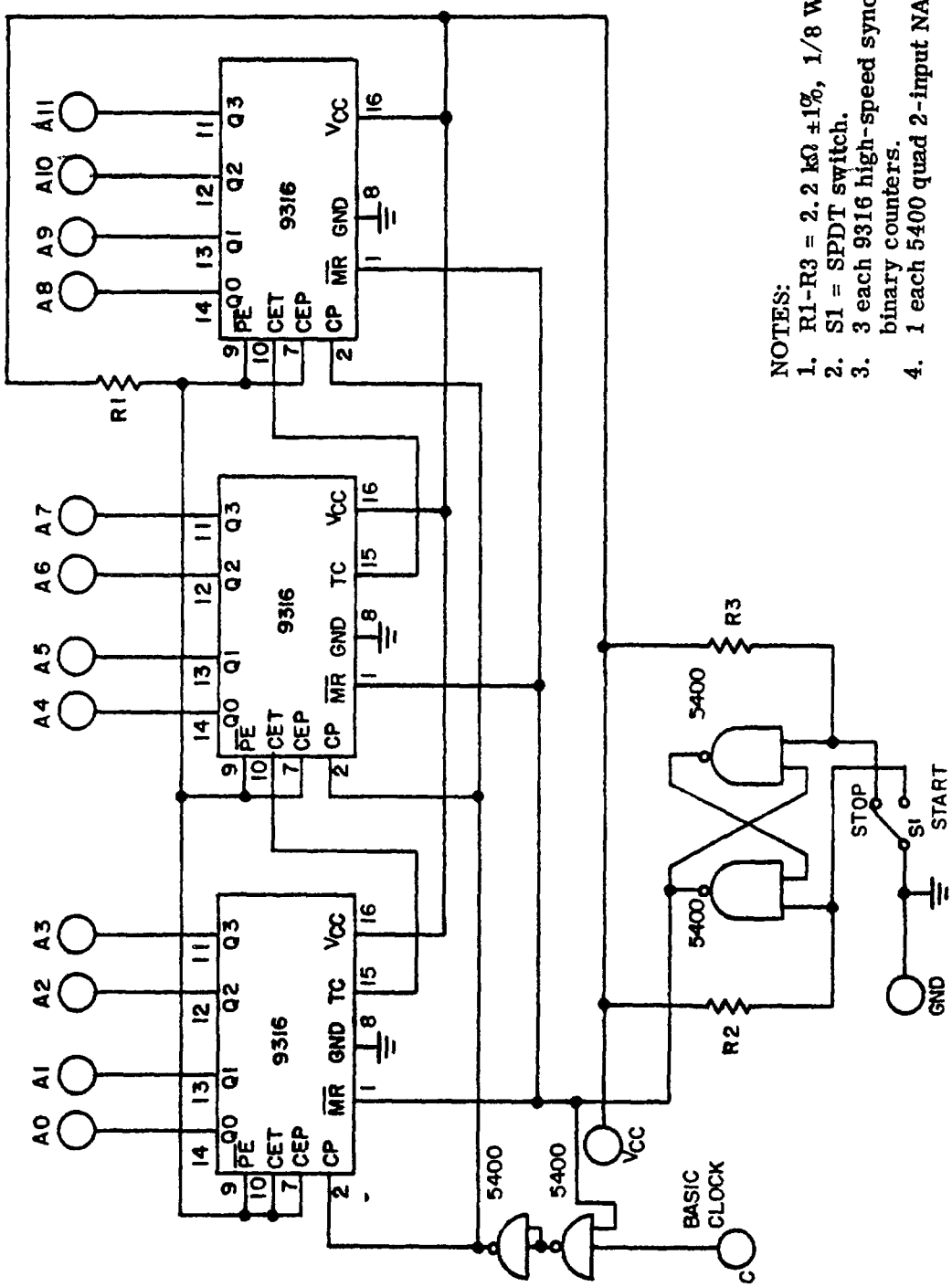
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NOTES:
1. R1-R4 = 499Ω ±1%, 1/4 W.
2. R5 = 51.1Ω ±5%, 1 W
3. Item J01 output parameters:
   PA = 4.0 ±1 V.
   PW = 500 ±50 ns.
   PRP = 1,000 ±100 ns.
   $t_{TLH} = t_{THL}$ = 25 ±20 ns.

* FIGURE 3. Operating temperature and life test circuit.

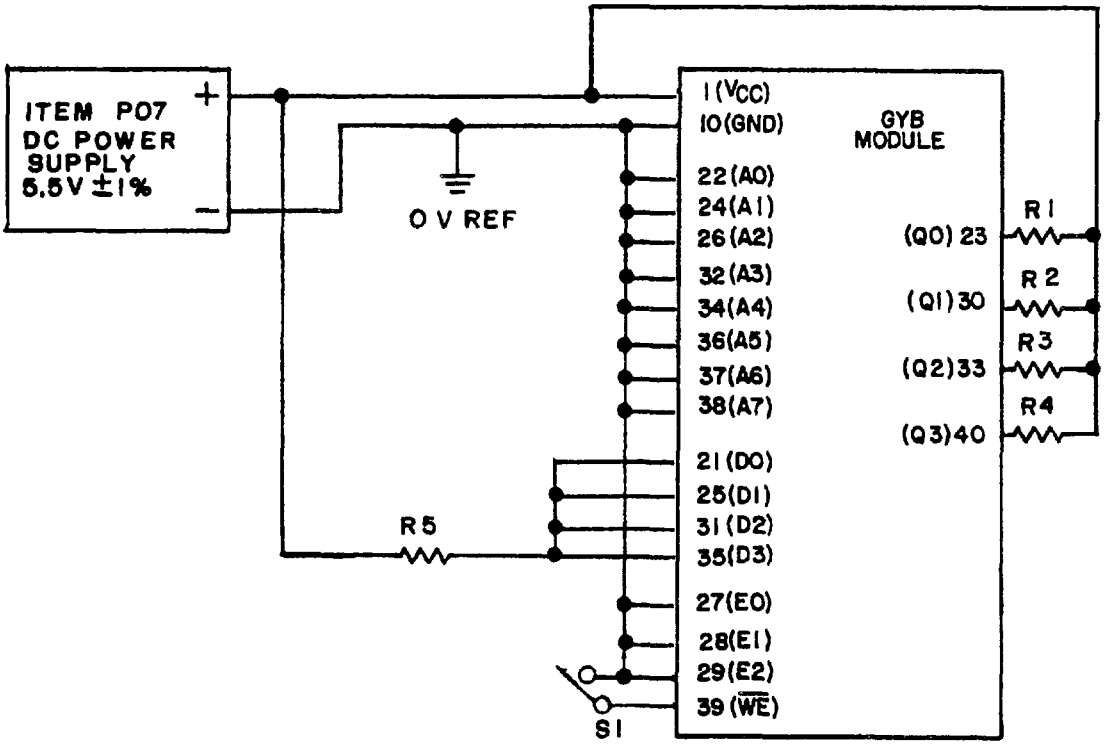
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NOTES:
1. R1-R3 = 2.2 kΩ ±1%, 1/8 W.
2. S1 = SPDT switch.
3. 3 each 9316 high-speed synchronous 4-bit binary counters.
4. 1 each 5400 quad 2-input NAND gate.

FIGURE 4. RAM address counter.

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NOTES:
1. R1-R4 = 499Ω ±1%, 1/4 W.
2. R5 = 51.1Ω ±5%, 1 W.
3. S1 = SPST switch.
4. Close switch S1 for a minimum of 500 ns, then open.
5. Verify that each output is at a low level (≤ 0.45 V).

*FIGURE 5. Component temperature test circuit.

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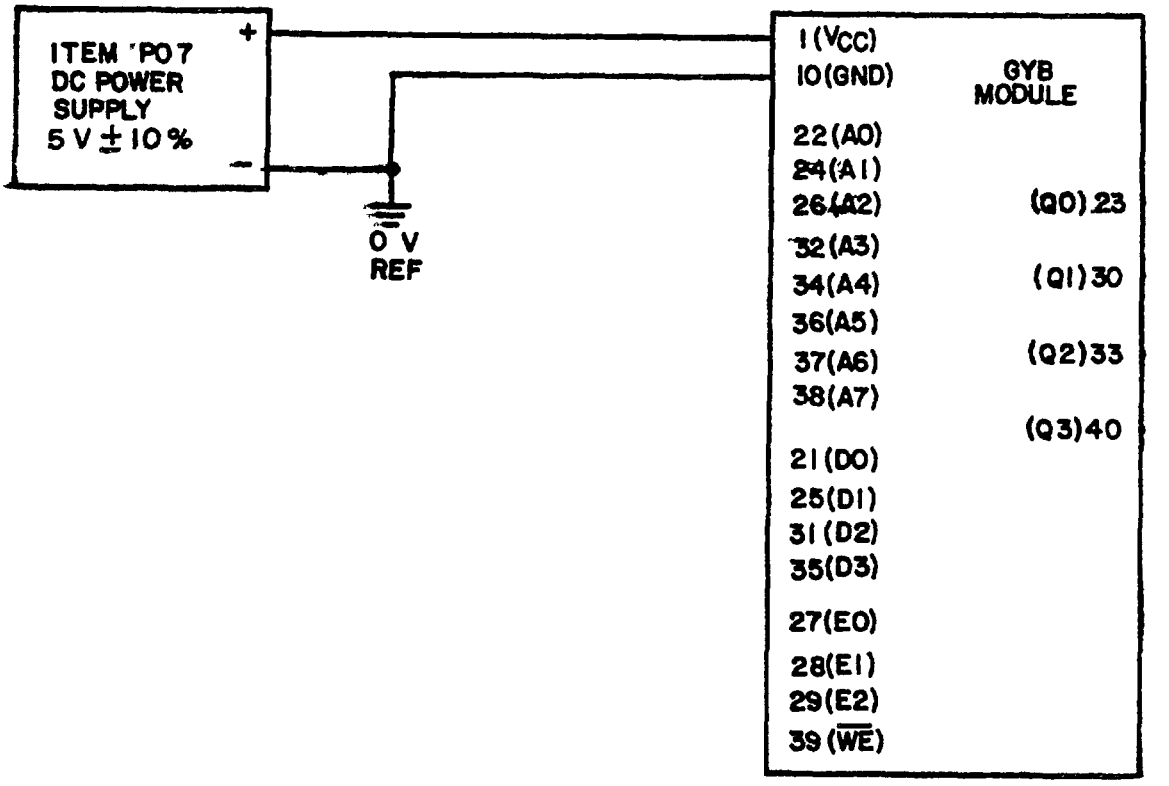


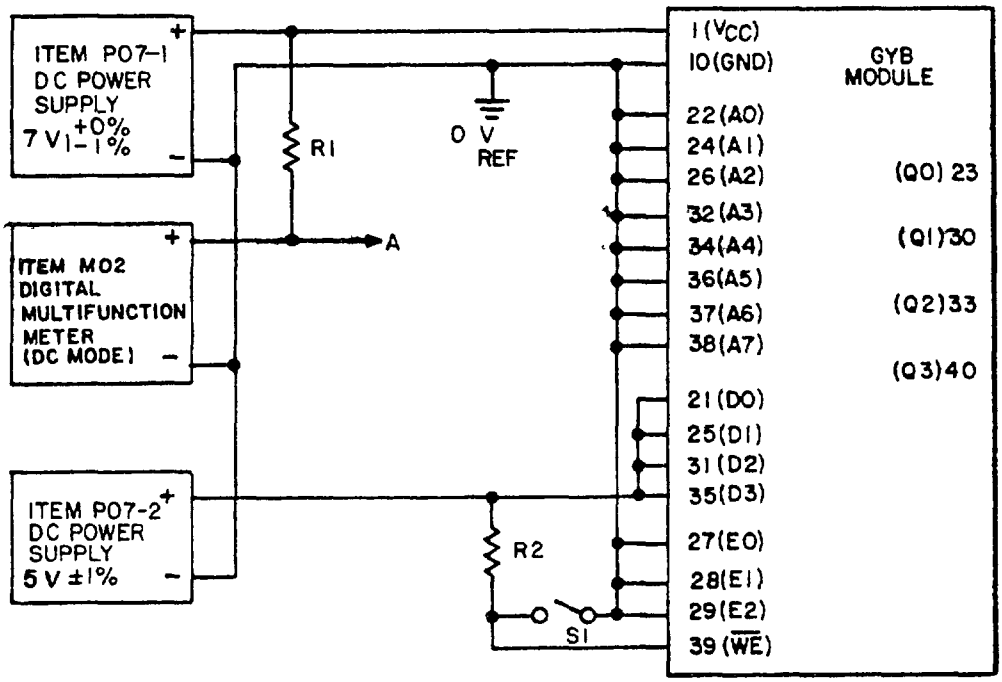
FIGURE 6. Vibration test circuit.

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TABLE II. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
A1	Q0 V_{OUT} (low loaded)	23	0	900	mV	7
A2	Q1 "	30	"	"	"	"
A3	Q2 "	33	"	"	"	"
A4	Q3 "	40	"	"	"	"

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NOTES:
1. R1 = 499Ω ±1%, 1/4 W.
2. R2 = 1.82 kΩ ±1%, 1/8 W
3. S1 = SPST switch.
4. Before making the connections at point A, close switch S1 for a minimum of 100 ns, then open and make point A connections.
5. Measurements shall be recorded after point A has been connected to the appropriate pin for a minimum of 1 second.

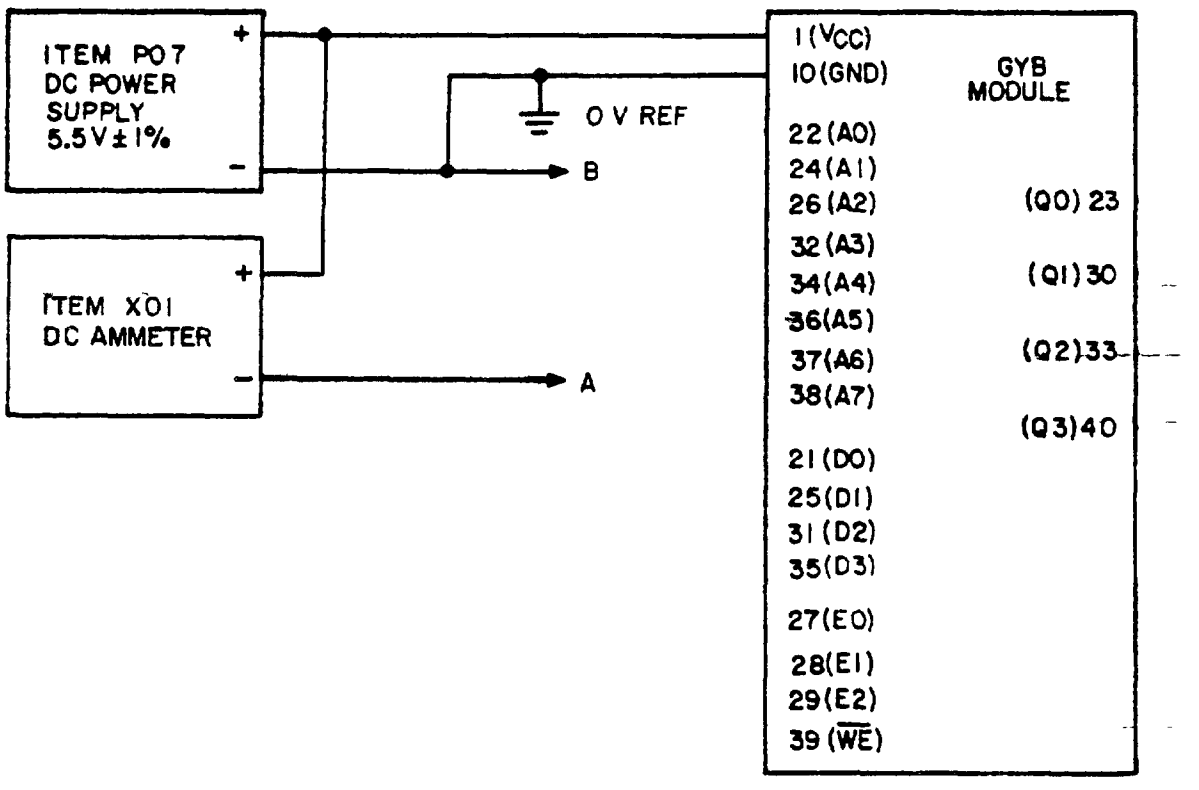
*FIGURE 7. V_{OUT} (low-loaded) test circuit for table II.

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TABLE III. Input breakdown current test.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
B1	A0	Input breakdown current	22	24,26,32,34,36,37,38,21,25,31,35,27,28,29,39	0	4	mA	8
B2	A1	"	24	22,26,32,34,36,37,38,21,25,31,35,27,28,29,39	"	"	"	"
B3	A2	"	26	22,24,32,34,36,37,38,21,25,31,35,27,28,29,39	"	"	"	"
B4	A3	"	32	22,24,26,34,36,37,38,21,25,31,35,27,28,29,39	"	"	"	"
B5	A4	"	34	22,24,26,32,36,37,38,21,25,31,35,27,28,29,39	"	"	"	"
B6	A5	"	36	22,24,26,32,34,37,38,21,25,31,35,27,28,29,39	"	"	"	"
B7	A6	"	37	22,24,26,32,34,36,38,21,25,31,35,27,28,29,39	"	"	"	"
B8	A7	"	38	22,24,26,32,34,36,37,21,25,31,35,27,28,29,39	"	"	"	"
B9	D1	"	21	22,24,26,32,34,36,37,38,25,31,35,27,28,29,39	"	1	"	"
B10	D2	"	25	22,24,26,32,34,36,37,38,21,31,35,27,28,29,39	"	"	"	"
B11	D3	"	31	22,24,26,32,34,36,37,38,21,25,35,27,28,29,39	"	"	"	"
B12	D4	"	35	22,24,26,32,34,36,37,38,21,25,31,27,28,29,39	"	"	"	"
B13	E0	"	27	22,24,26,32,34,36,37,38,21,25,31,35,28,29,39	"	4	"	"
B14	E1	"	28	22,24,26,32,34,36,37,38,21,25,31,35,27,29,39	"	"	"	"
B15	E2	"	29	22,24,26,32,34,36,37,38,21,25,31,35,27,28,39	"	"	"	"
B16	$\overline{WE}$	"	39	22,24,26,32,34,36,37,38,21,25,31,35,27,28,29	"	"	"	"

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CAUTION: Do not isolate the chassis of item X01 from earth ground.

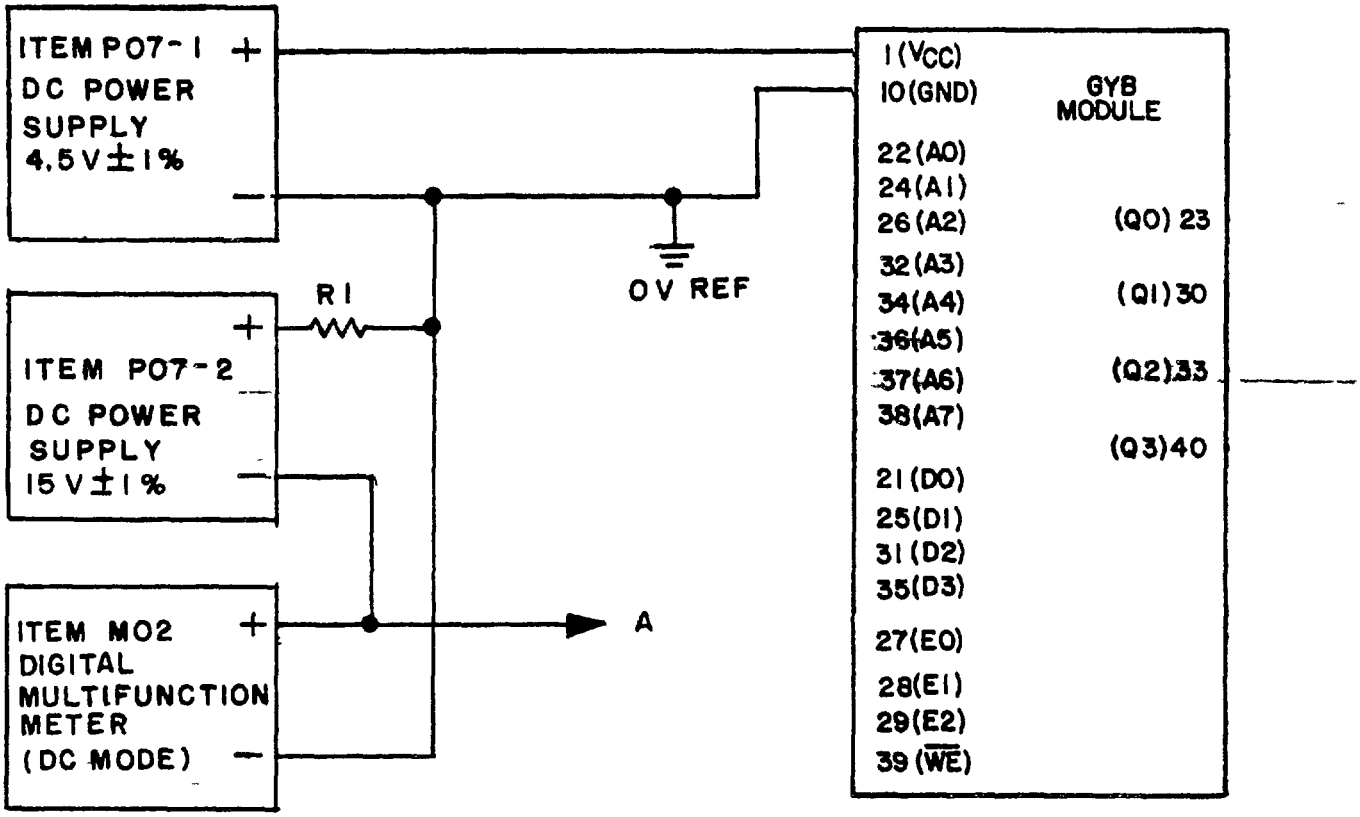
FIGURE 8. Input breakdown current test circuit for table III.

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TABLE IV. Input clamp diode voltage test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
C1	A0 Input clamp diode voltage	22	-0.2	-1.5	V	9
C2	A1 "	24	"	"	"	"
C3	A2 "	26	"	"	"	"
C4	A3 "	32	"	"	"	"
C5	A4 "	34	"	"	"	"
C6	A5 "	36	"	"	"	"
C7	A6 "	37	"	"	"	"
C8	A7 "	38	"	"	"	"
C9	D0 "	21	"	"	"	"
C10	D1 "	25	"	"	"	"
C11	D2 "	31	"	"	"	"
C12	D3 "	35	"	"	"	"
C13	E0 "	27	"	"	"	"
C14	E1 "	28	"	"	"	"
C15	E2 "	29	"	"	"	"
C16	$\overline{WE}$ "	39	"	"	"	"

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NOTE: R1 = 1.82 kΩ ±1%, 1/8 W.

* FIGURE 9. Input clamp diode voltage test circuit for table IV.

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*TABLE V. Isolation test.

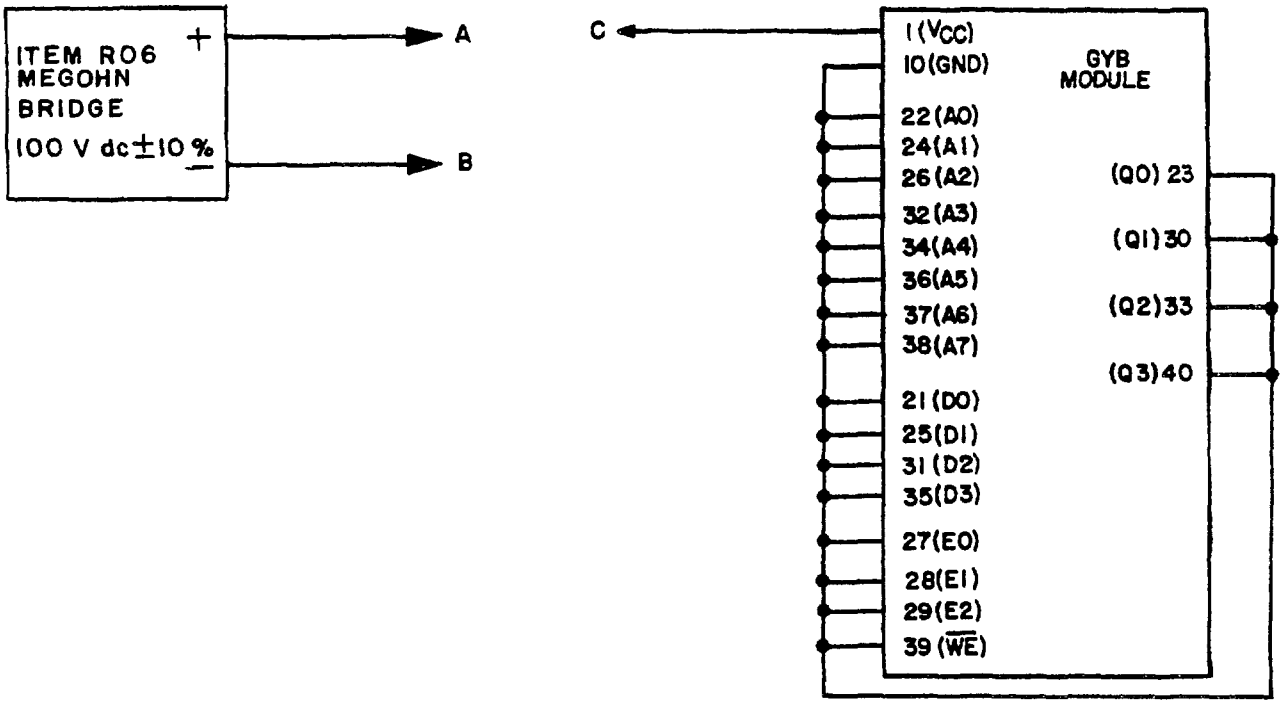
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
D1	Isolation, frame to module pins	C	Frame	10	∞	MΩ	10
D2	Isolation, used to unused pins	C	2,3,4,5,6,7,8,9,11,12,13,14,15,16, 17,18,19,20	"	"	"	"
D3	Isolation, unused to unused pins	2	3,4,5,6,7,8,9,11,12,13,14,15,16, 17,18,19,20	"	"	"	"
D4	"	3	2,4,5,6,7,8,9,11,12,13,14,15,16, 17,18,19,20	"	"	"	"
D5	"	4	2,3,5,6,7,8,9,11,12,13,14,15,16, 17,18,19,20	"	"	"	"
D6	"	5	2,3,4,6,7,8,9,11,12,13,14,15,16, 17,18,19,20	"	"	"	"
D7	"	6	2,3,4,5,7,8,9,11,12,13,14,15,16, 17,18,19,20	"	"	"	"
D8	"	7	2,3,4,5,6,8,9,11,12,13,14,15,16, 17,18,19,20	"	"	"	"
D9	"	8	2,3,4,5,6,7,9,11,12,13,14,15,16, 17,18,19,20	"	"	"	"
D10	"	9	2,3,4,5,6,7,8,11,12,13,14,15,16, 17,18,19,20	"	"	"	"
D11	"	11	2,3,4,5,6,7,8,9,12,13,14,15,16, 17,18,19,20	"	"	"	"
D12	"	12	2,3,4,5,6,7,8,9,11,13,14,15,16, 17,18,19,20	"	"	"	"
D13	"	13	2,3,4,5,6,7,8,9,11,12,14,15,16, 17,18,19,20	"	"	"	"
D14	"	14	2,3,4,5,6,7,8,9,11,12,13,15,16, 17,18,19,20	"	"	"	"
D15	"	15	2,3,4,5,6,7,8,9,11,12,13,14,16, 17,18,19,20	"	"	"	"

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*TABLE V. Isolation test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
D16	Isolation, unused to unused pins	16	2,3,4,5,6,7,8,9,11,12,13,14,15,17,18,19,20	10	∞	MΩ	10
D17	"	17	2,3,4,5,6,7,8,9,11,12,13,14,15,16,18,19,20	"	"	"	"
D18	"	18	2,3,4,5,6,7,8,9,11,12,13,14,15,16,17,19,20	"	"	"	"
D19	"	19	2,3,4,5,6,7,8,9,11,12,13,14,15,16,17,18,20	"	"	"	"
D20	"	20	2,3,4,5,6,7,8,9,11,12,13,14,15,16,17,18,19	"	"	"	"

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WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: On test condition D1 (when measuring to frame), penetrate the anodized finish of the module frame with a sharply pointed probe connected to the negative terminal.

*FIGURE 10. Isolation test circuit for table V.

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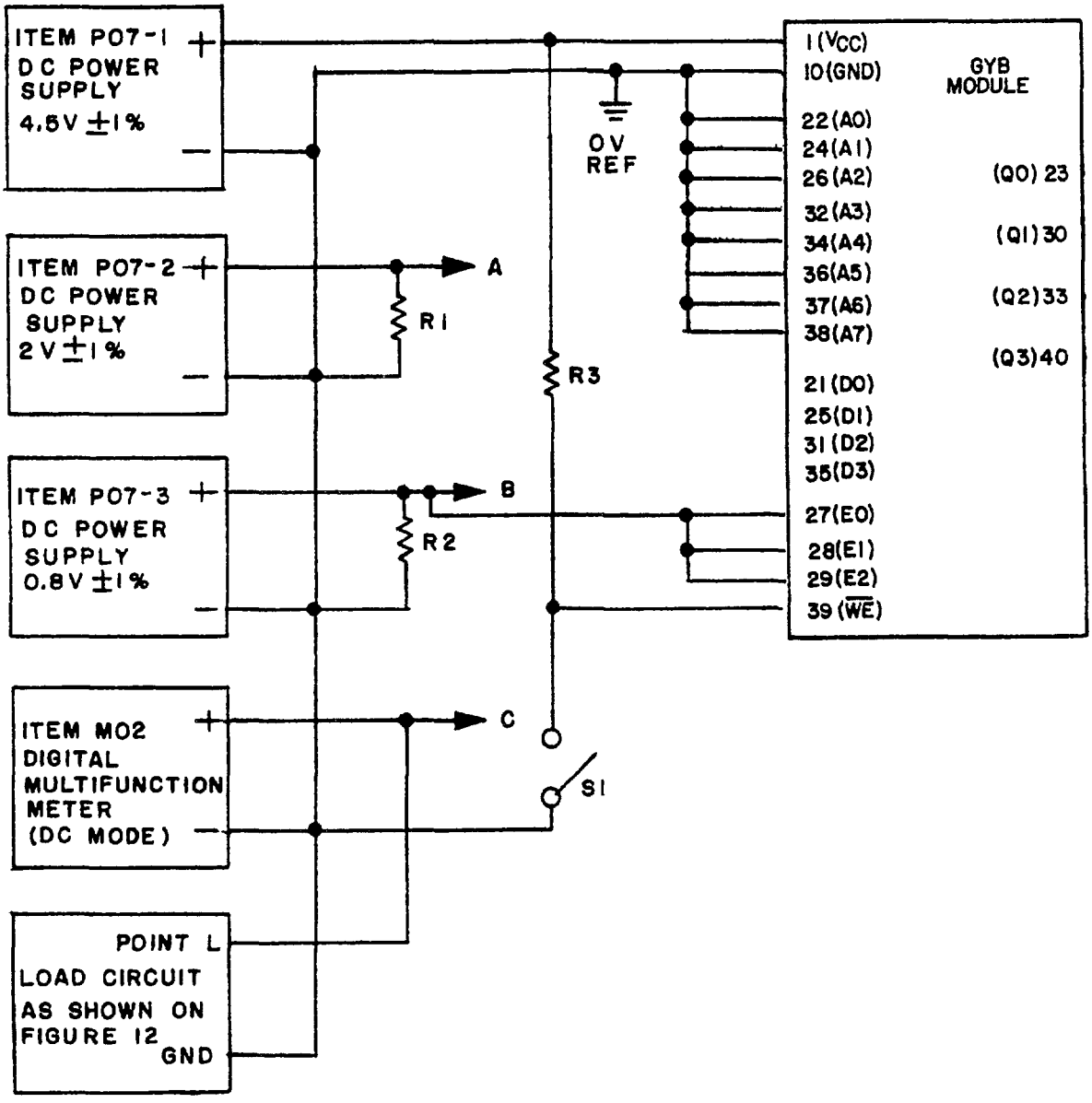
*TABLE VI. Output voltage (low-level) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
E1	Q0 Output voltage, low level	21,25,31,35	Open	23	0	450	mV	11
E2	Q1 "	"	"	30	"	"	"	"
E3	Q2 "	"	"	33	"	"	"	"
E4	Q3 "	"	"	40	"	"	"	"

*TABLE VII. Output voltage (high-level) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
F1	Q0 Output voltage, high level	Open	21,25,31,35	23	2.4	4.5	V	11
F2	Q1 "	"	"	30	"	"	"	"
F3	Q2 "	"	"	33	"	"	"	"
F4	Q3 "	"	"	40	"	"	"	"

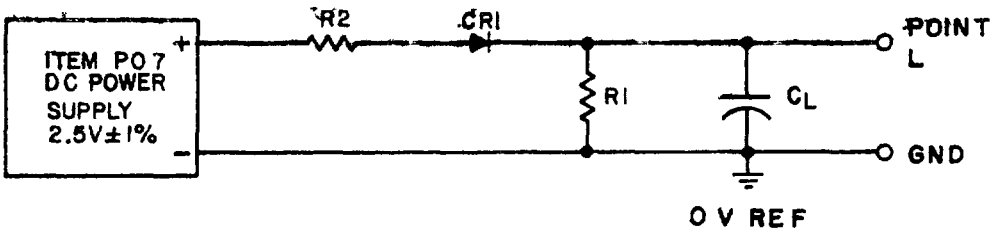
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:16Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = R2 = 51.1Ω ±5%, 1 W.
2. R3 = 1.82 kΩ ±1%, 1/8 W.
3. S1 = SPST switch.
4. Before making measurement, close switch S1 for a minimum of 100 ns, then open it and make measurement.

*FIGURE 11. Output voltage (low-level and high-level) test circuit for tables VI and VII.

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NOTES:
1. R1 = 750Ω ±1%, 1/8 W (current flow of 3.2 mA when point L is held at 2.4 V).
2. R2 = Select value for a current flow of 10 mA ±5% out of point L when point L is held at 0.45 V.
3. CR1 = 1N916, or equivalent.
4. For test conditions M1-M4, N1-N4, R1, R2, S1, S2, T1, T2, U1, U2, and W1-W8, C_L = 50 ±5 pF (including probe capacitance).
5. For test conditions M5-M8, N5-N8, R3, R4, S3, S4, T3, T4, U3 and U4, C_L = 150 ±5 pF (including probe capacitance).
6. For test conditions Q1-Q4 and P1-P4, $C_L \leq 10$ ±2 pF (including probe capacitance.
7. C_L is not required for dc electrical test conditions E, F, and G.
8. Lead length from load circuit to module output pin under test is to be less than 2 inches (50.80 mm).

*FIGURE 12. GYB logic load circuit.

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*TABLE VIII. Address threshold (low-level and high-level) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G1	Address threshold, low level	39	22,24,26,32,34, 36,37,38,21,25, 31,35,27,28,29	23	0	450	mV	13
G2	"	"	"	30	"	"	"	"
G3	"	"	"	33	"	"	"	"
G4	"	"	"	40	"	"	"	"
G5	Address threshold, high level	39,22,24,26,32, 34,36,37,38,21, 25,31,35	27,28,29	23	2.4	4.5	V	"
G6	"	"	"	30	"	"	"	"
G7	"	"	"	33	"	"	"	"
G8	"	"	"	40	"	"	"	"

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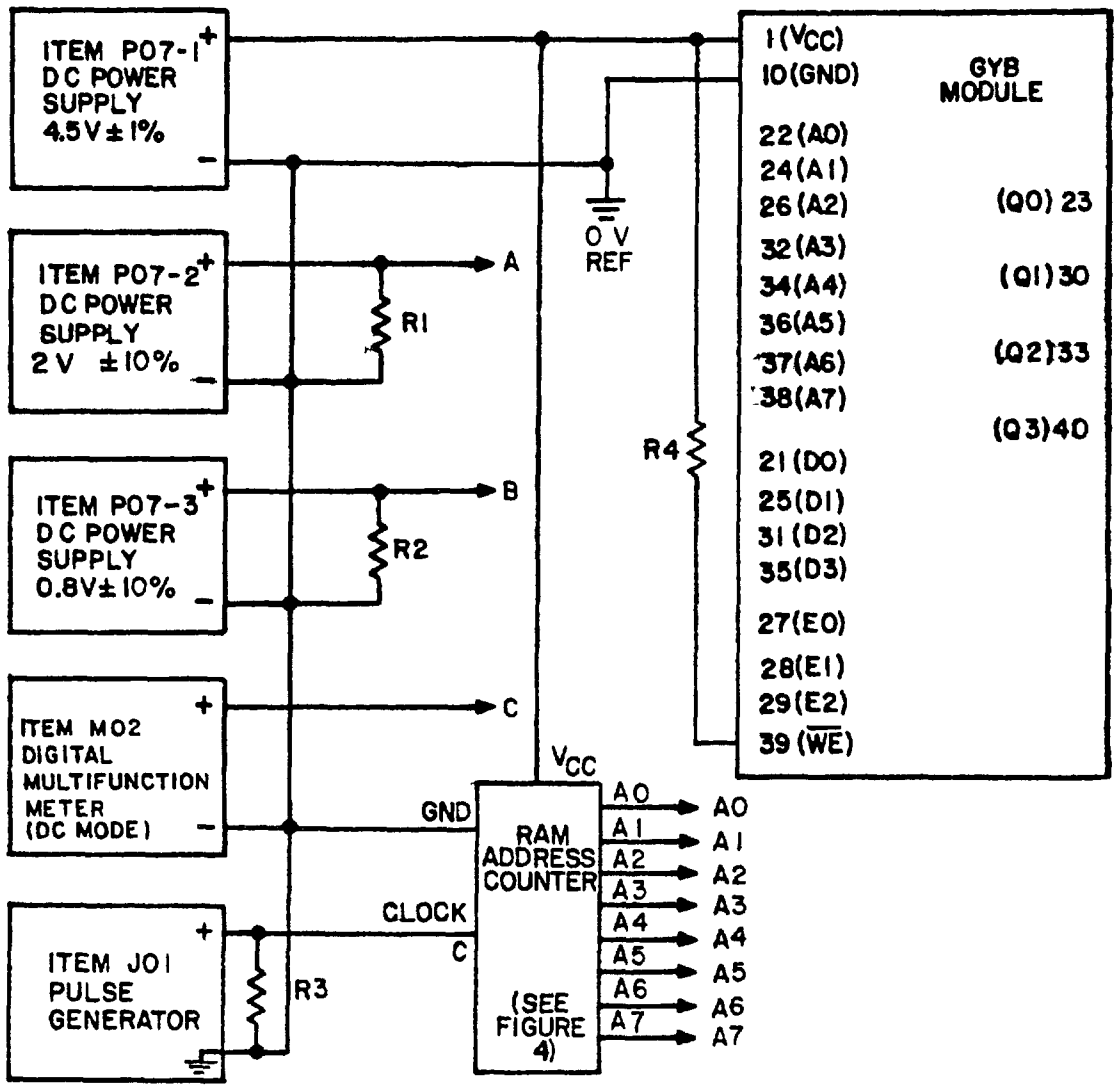


*FIGURE 13. Address threshold (low-level and high-level) test circuit for table VIII.

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NOTES:
1. R1 = R2 = R3 = 51.1Ω ±5%, 1 W.
2. R4 = 1.82 kΩ ±1%, 1/4 W.
3. Load all outputs with the load as shown on figure 12.
4. Item J01 output parameters:

   PA = 4 ±1 V.
   PW = 500 ±50 ns.
   PRR = 1 MHz ±10%.
   $t_{TLH} = t_{THL}$ = 25 ±20 ns.

5. Test sequence for test conditions G1-G4:

   a. Connect the RAM address counter to address inputs.
   b. Connect pins 27, 28, 29, 39, 21, 25, 31, and 35 to pin 10.
   c. Initiate RAM address counter for 1 second which will load all memory locations with a high level.
   d. Disconnect pin 39 from pin 10.
   e. Disconnect the RAM address counter and address location zero by connecting pins 22, 24, 26, 32, 34, 36, 37, and 38 to pin 10. Connect pins 21, 25, 31, and 35 to pin 1.
   f. Connect pin 39 to pin 10 for a minimum of 100 ns.
   g. Address location zero should have a low level stored and all other locations should have a high level stored.
   h. Make connections as specified in table VIII for test conditions G1-G4 without interrupting power. The connection of pin 39 to point A should be the last connection.

6. Test sequence for test conditions G5-G8:

   a. Connect the RAM address counter to address inputs.
   b. Connect pins 27, 28, 29, and 39 to pin 10. Connect pins 21, 25, 31, and 35 to pin 1.
   c. Initiate RAM address counter for 1 second which will load all memory locations with a low level.
   d. Disconnect pin 39 from pin 10.
   e. Disconnect the RAM address counter and address location 256 by connecting pins 22, 24, 26, 32, 34, 36, 37, and 38 to pin 1. Connect pins 21, 25, 31, and 35 to pin 10.
   f. Connect pin 39 to pin 10 for a minimum of 100 ns.
   g. Address location 256 should have a high level stored and all other locations should have a low level stored.
   h. Make connections as specified in table VIII for test conditions G5-G8 without interrupting power. The connection of pin 39 to point A should be the last connection.

7. Notes 5g and 6g must be verified. This verification is important to detect modules which will not write a high or a low.

*FIGURE 13. Address threshold (low-level and high-level) test circuit for table VIII - Continued.

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TABLE IX. Input current (low-level) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
H1	A0 Input current, low level	Open	Open	22	24,26,32,34,36,37,38	0.0	3.5	mA	14
H2	A1 "	"	"	24	22,26,32,34,36,37,38	"	"	"	"
H3	A2 "	"	"	26	22,24,32,34,36,37,38	"	"	"	"
H4	A3 "	"	"	32	22,24,26,34,36,37,38	"	"	"	"
H5	A4 "	"	"	34	22,24,26,32,36,37,38	"	"	"	"
H6	A5 "	"	"	36	22,24,26,32,34,37,38	"	"	"	"
H7	A6 "	"	"	37	22,24,26,32,34,36,38	"	"	"	"
H8	A7 "	"	"	38	22,24,26,32,34,36,37	"	"	"	"
H9	D0 "	"	"	21	25,31,35	0.00	0.87	"	"
H10	D1 "	"	"	25	21,31,35	"	"	"	"
H11	D2 "	"	"	31	21,25,35	"	"	"	"
H12	D3 "	"	"	35	21,25,31	"	"	"	"
H13	E0 "	"	"	27	28,29	0.0	3.5	"	"
H14	E1 "	"	"	28	27,29	"	"	"	"
H15	E2 "	"	"	29	27,28	"	"	"	"
H16	$\overline{WE}$ "	"	"	39	Open	"	"	"	"

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TABLE X. Input current (high-level) test.

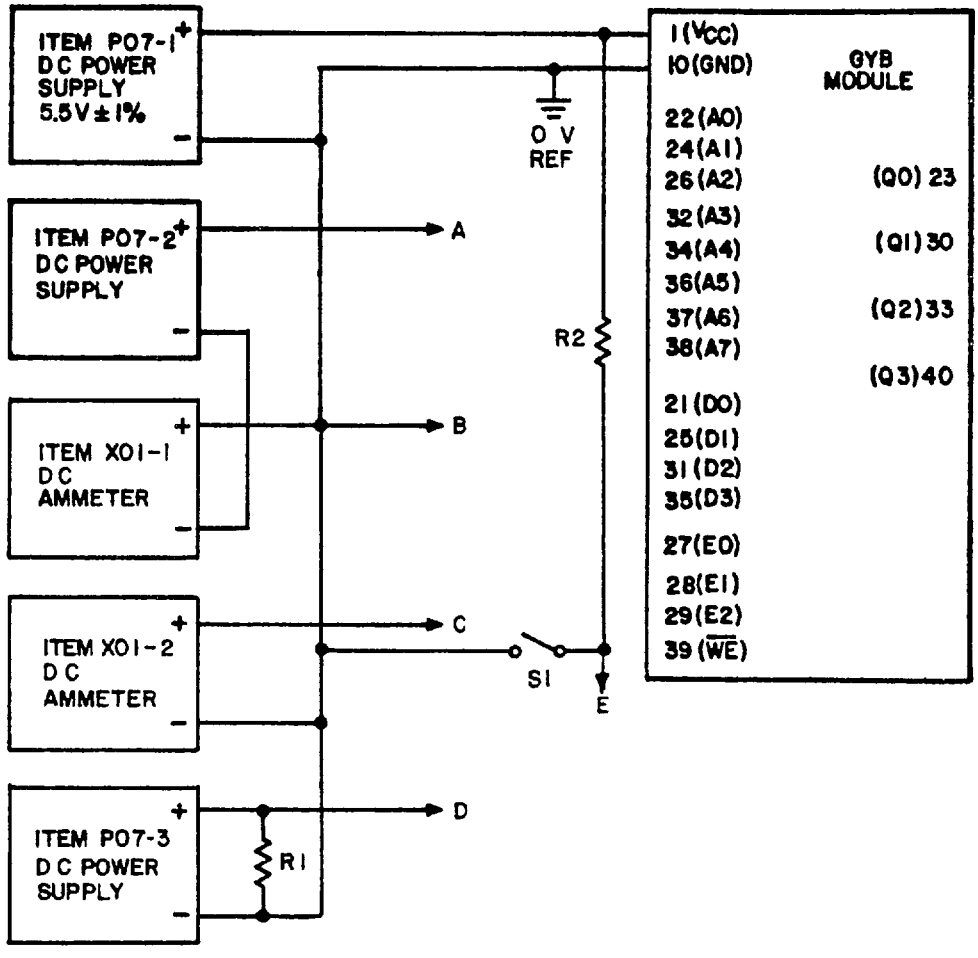
Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
J1	A0 Input current, high level	22	24,26,32,34,36,37,38	Open	Open	-1,600	100	µA	14
J2	A1 "	24	22,26,32,34,36,37,38	"	"	"	"	"	"
J3	A2 "	26	22,24,32,34,36,37,38	"	"	"	"	"	"
J4	A3 "	32	22,24,26,34,36,37,38	"	"	"	"	"	"
J5	A4 "	34	22,24,26,32,36,37,38	"	"	"	"	"	"
J6	A5 "	36	22,24,26,32,34,37,38	"	"	"	"	"	"
J7	A6 "	37	22,24,26,32,34,36,38	"	"	"	"	"	"
J8	A7 "	38	22,24,26,32,34,36,37	"	"	"	"	"	"
J9	D0 "	21	25,31,35	"	"	-400	25	"	"
J10	D1 "	25	21,31,35	"	"	"	"	"	"
J11	D2 "	31	21,25,35	"	"	"	"	"	"
J12	D3 "	35	21,25,31	"	"	"	"	"	"
J13	E0 "	27	28,29	"	"	-1,600	100	"	"
J14	E1 "	28	27,29	"	"	"	"	"	"
J15	E2 "	29	27,28	"	"	"	"	"	"
J16	$\overline{WE}$ "	39	Open	"	"	"	"	"	"

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TABLE XI. Output short circuit current and output leakage current test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
K1	Q0 Output short circuit current	Open	21,22,24,25,26,27,28,29, 31,32,34,35,36,37,38	23	Open	20	100	mA	14
K2	Q1 "	"	"	30	"	"	"	"	"
K3	Q2 "	"	"	33	"	"	"	"	"
K4	Q3 "	"	"	40	"	"	"	"	"
K5	Q0 Output leakage current (V_{OUT} = 450 mV)	23	21,25,31,35	Open	27,28,29	-100	"	µA	"
K6	Q1 "	30	"	"	"	"	"	"	"
K7	Q2 "	33	"	"	"	"	"	"	"
K8	Q3 "	40	"	"	"	"	"	"	"
K9	Q0 Output leakage current (V_{OUT} = 5.5 V)	23	Open	"	"	0	"	"	"
K10	Q1 "	30	"	"	"	"	"	"	"
K11	Q2 "	33	"	"	"	"	"	"	"
K12	Q3 "	40	"	"	"	"	"	"	"

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CAUTION: Do not isolate the chassis of item X01 from earth ground.

NOTES:
1. R1 = 51.1Ω ±5%, 1 W.
2. R2 = 1.82 kΩ ±1%, 1/8 W.
3. Since items X01-1 and X01-2 are not used simultaneously, only one item X01 is required. Use item X01-2 for tables IX and XI; use item X01-1 for table X.
4. With point "A" disconnected, item X01-1 should indicate zero current on the 1 µA range.
5. For test condition H, set item P07-3 to 4.5 V ±1% and make measurement.
6. For test condition J, set item P07-2 to 2.4 V ±1% and make measurement.
7. For test conditions K1-K4, connect point E to pin 39. Close switch S1 for a minimum of 100 ns, then open. When performing test conditions K1-K4, only one output shall be shorted at a time and the measurement taken within 5 seconds.
8. For test conditions K5-K8, set item P07-2 to 450 mV ±1% and item P07-3 to 2 V ±1%. Set item P07-2 current limit at 1 mA.
9. For test conditions K9-K12, set item P07-2 to 5.5 V ±1%, and item P07-3 to 2 V ±1%. Set item P07-2 current limit at 1 mA.
10. Item P07-2 must be capable of sinking current.

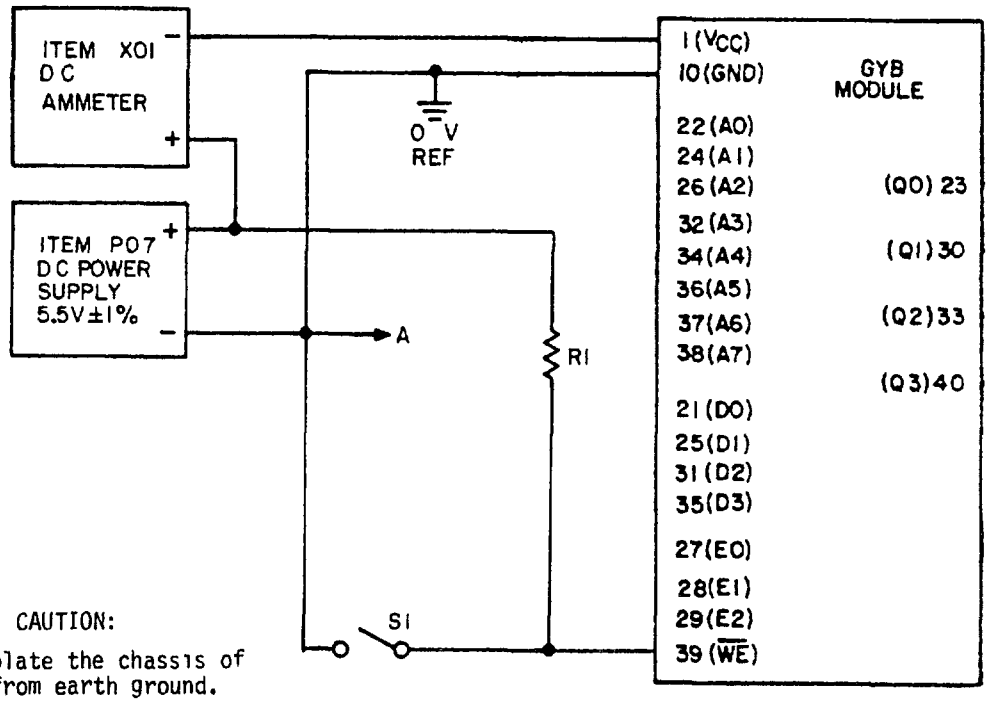
* FIGURE 14. Input current (low-level and high-level), output short circuit current, and output leakage current test circuit for tables IX, X, and XI.

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TABLE XII. Power supply current test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
L1	Power supply current, low level	21,25,27,28,29,31,35	5	455	mA	15
L2	Power supply current, high level	27,28,29	"	"	"	"
L3	Power supply current, high-impedance state	Open	"	"	"	"

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CAUTION:
Do not isolate the chassis of item XOI from earth ground.

NOTES:
1. RI = 1.82 kΩ ±1%, 1/8 W.
2. SI = SPST switch.
3. For test conditions L1 and L2: Close SI for a minimum of 100 ns, then open and make measurement.
4. For test condition L3, SI is open.

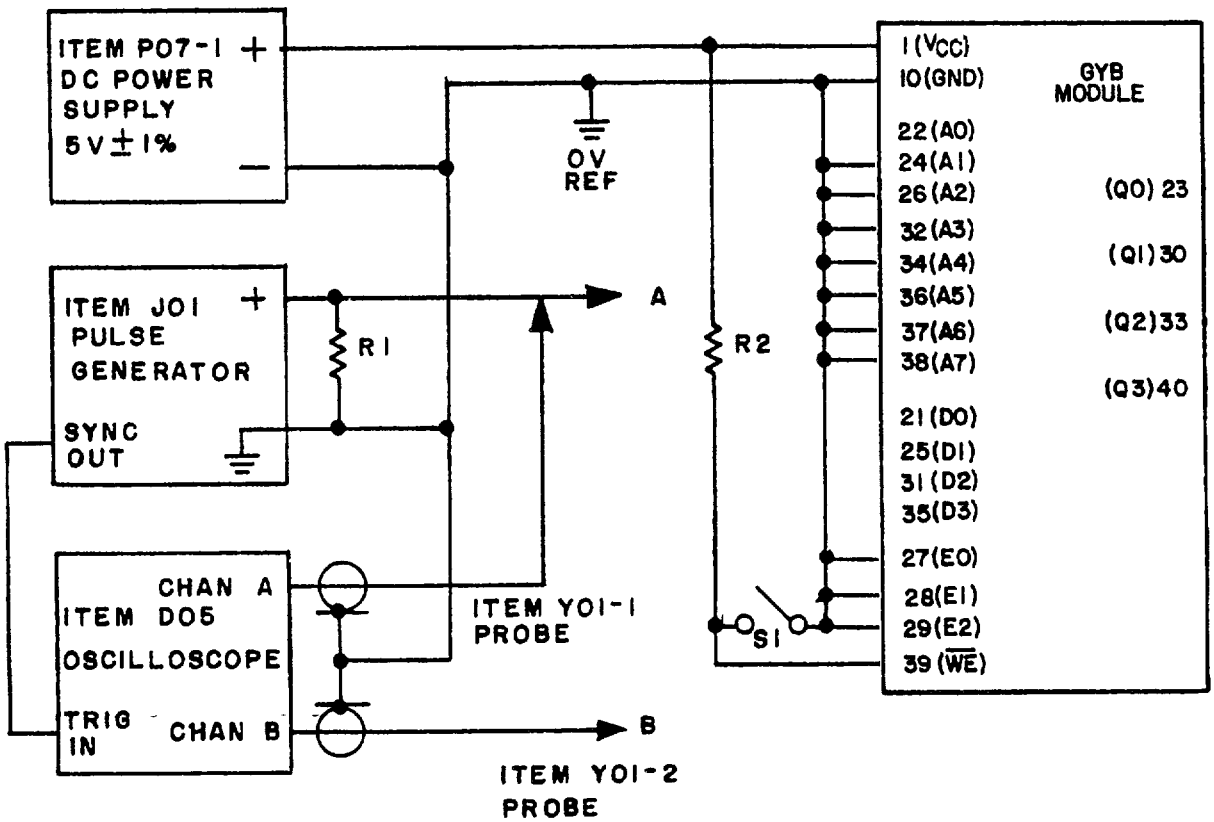
FIGURE 15. Power supply current test circuit for table XII.

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*TABLE XIII. Transition time (t_{TLH} and t_{THL}) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
M1	Transition time (t_{TLH}), C_L = 50 pF	22	23	5	25	ns	16,17
M2	"	"	30	"	"	"	"
M3	"	"	33	"	"	"	"
M4	"	"	40	"	"	"	"
M5	Transition time (t_{TLH}), C_L = 150 pF	"	23	"	35	"	"
M6	"	"	30	"	"	"	"
M7	"	"	33	"	"	"	"
M8	"	"	40	"	"	"	"
N1	Transition time (t_{THL}), C_L = 50 pF	"	23	2	15	"	"
N2	"	"	30	"	"	"	"
N3	"	"	33	"	"	"	"
N4	"	"	40	"	"	"	"
N5	Transition time (t_{THL}), C_L = 150 pF	"	23	3	25	"	"
N6	"	"	30	"	"	"	"
N7	"	"	33	"	"	"	"
N8	"	"	40	"	"	"	"

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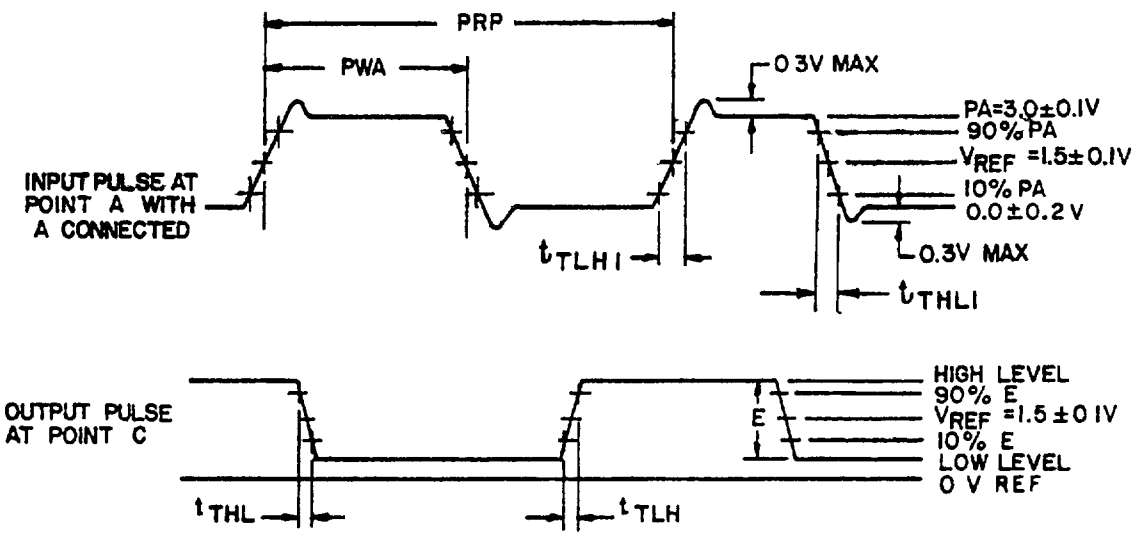
NOTES:
1. R1 = 51.1Ω ±5%, 1 W.
2. R2 = 1.82 kΩ ±1%, 1/8 W.
3. S1 = SPST switch.
4. Load all outputs with the load shown on figure 12.
5. Initial memory loading procedure:

   a. Connect pins 22, 21, 25, 31, and 35 to pin 10 and close switch S1 for a minimum of 100 ns, then open. Memory location zero is now loaded with a high level.

   b. Connect pins 22, 21, 25, 31, and 35 to pin 1 and close switch S1 for a minimum of 100 ns, then open and do not close switch S1 again for the remainder of testing. Memory location one is now loaded with a low level.

*FIGURE 16. Transition time (t_{TLH} and t_{THL}) test circuit for table XIII.

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NOTES:
1. Input pulse parameters:
   $t_{TLH1} = t_{THL1} = 10 \pm 2$ ns
   PWA = 500 ±50 ns
   PRP = 1,000 ±100 ns
2. Before making the measurements, verify that the output pulse high level is $\geq$2.4 V and the low level is $\leq$0.45 V.
3. All voltages are referenced to ground.

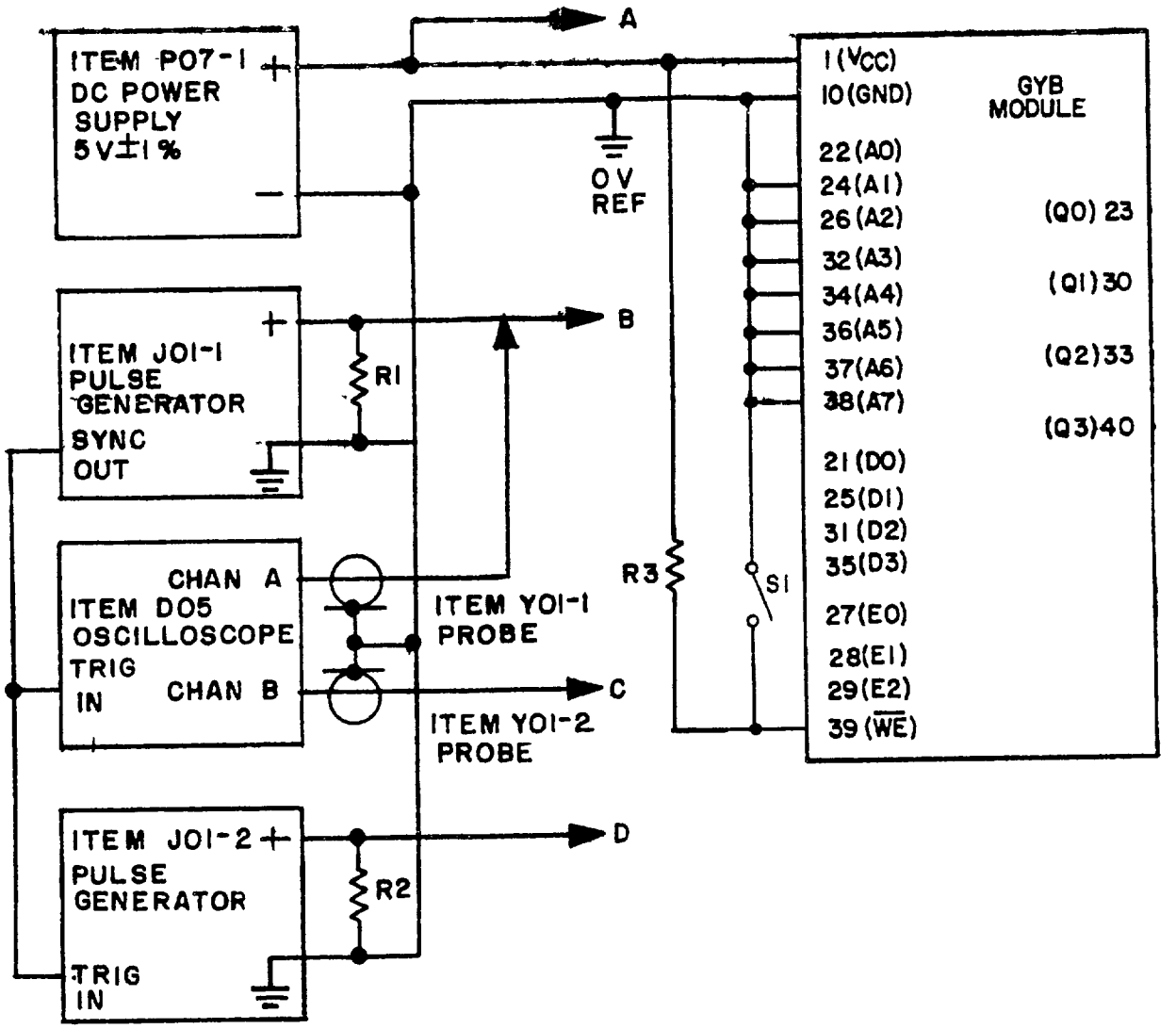
*FIGURE 17. Transition time (t_{TLH} and t_{THL}) test circuit waveforms for table XIII.

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TABLE XIV. Enable recovery time (low-state to high-impedance state (t_{ERL}), high-state to high-impedance state (t_{ERH})) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P1	Enable recovery time, low-state to high-impedance state (t_{ERL}), Q0 output	21,25,31,35	27,28,29	23	22	1	40	ns	18,19
P2	Enable recovery time, low-state to high-impedance state (t_{ERL}), Q1 output	"	"	30	"	"	"	"	"
P3	Enable recovery time, low-state to high-impedance state (t_{ERL}), Q2 output	"	"	33	"	"	"	"	"
P4	Enable recovery time, low-state to high-impedance state (t_{ERL}), Q3 output	"	"	40	"	"	"	"	"
Q1	Enable recovery time, high-state to high-impedance state (t_{ERH}), Q0 output	"	"	23	"	"	"	"	"
Q2	Enable recovery time, high-state to high-impedance state (t_{ERH}), Q1 output	"	"	30	"	"	"	"	"
Q3	Enable recovery time, high-state to high-impedance state (t_{ERH}), Q2 output	"	"	33	"	"	"	"	"
Q4	Enable recovery time, high-state to high-impedance state (t_{ERH}), Q3 output	"	"	40	"	"	"	"	"

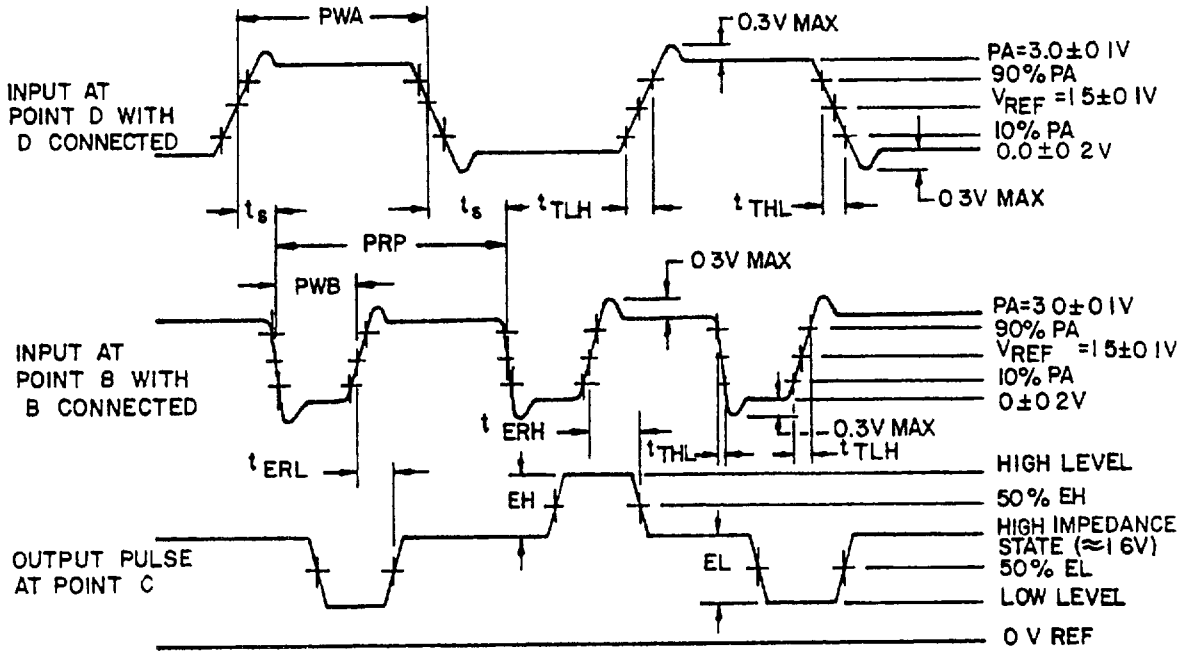
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NOTES:
1. R1 = R2 = 51.1Ω ±5%, 1 W.
2. R3 = 1.82 kΩ ±1%, 1/8 W.
3. S1 = SPST switch.
4. Load all outputs with the load shown on figure 12.

*FIGURE 18. Enable recovery time (low-state to high-impedance state (t_{ERL}), high-state to high-impedance state (t_{ERH}) test circuit for table XIV.

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NOTES:
1. Input pulse parameters:
   PWA = 500 ±50 ns.
   PWB = 100 ±10 ns.
   PRP = 500 ±50 ns.
   t_S = 200 ±20 ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. Initial setup procedure:

   a. Store a high level in address location zero by connecting pins 22, 21, 25, 31, 35, 27, 28, and 29 to pin 10. Close switch S1 for a minimum of 100 ns, then open.

   b. Store a low level in address location one by connecting pins 22, 21, 25, 31, and 35 to pin 1 and connecting pins 27, 28, and 29 to pin 10. Close switch S1 for a minimum of 100 ns, then open.

   c. Verify that address location zero has a high-level stored and address location one has a low-level stored.

* FIGURE 19. Enable recovery time (low-state to high-impedance state (t_{ERL}), high-state to high-impedance state (t_{ERH})) test circuit waveforms for table XIV.

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TABLE XV. Walking-one-walking-zero test pattern test.

Test condition	Parameter		Initial limits 25°C ±5°C ambient		Test figure
			Min	Max	
R1	Walking-one-walking-zero test pattern (C_L = 50 pF, V_{CC} = 4.5 V)	1/, 2/, 3/, 4/, 6/	1/ - 7/		20,21, 22,23
R2	Walking-one-walking-zero test pattern (C_L = 50 pF, V_{CC} = 5.5 V)	1/, 2/, 3/, 4/, 7/	"		"
R3	Walking-one-walking-zero test pattern (C_L = 150 pF, V_{CC} = 4.5 V)	1/, 2/, 3/, 5/, 6/	"		"
R4	Walking-one-walking-zero test pattern (C_L = 150 pF, V_{CC} = 5.5 V)	1/, 2/, 3/, 5/, 7/	"		"

1/ Walking-one-walking-zero pattern testing. The objective of walking-one-walking-zero pattern testing (writing a one in a field of zeros and a zero in a field of ones) is to show the independence of each bit in both the zero and one state with respect to all others bits which are in the complement state. This pattern also verifies minimum write pulse width (t_{WP}), address setup time (t_{AS}), data setup time (t_{DS}), enable setup time (t_{ES}), data hold time (t_{DH}), enable hold time (t_{EH}), address access time (t_{AA}), and enable access time (t_{EA}).

2/ A program that fulfills the walking-one-test pattern would be as follows:

(a) Write "lows" in all memory address locations.
(b) Read "lows" in all memory address locations.
(c) Write a "high" in memory output "Q0" at address location "0".
(d) Read a "low" in all other memory bit locations.
(e) Read a "high" in memory output "Q0" at address location "0".
(f) Write a "low" in memory output "Q0" at address location "0".
(g) Write a "high" in memory output "Q1" at address location "0".
(h) Read a "low" in all other memory bit locations.
(i) Read a "high" in memory output "Q1" at address location "0".
(j) Write a "low" in memory output "Q1" at address location "0".
(k) Write a "high" in memory output "Q2" at address location "0".
(l) Read a "low" in all other memory bit locations.
(m) Read a "high" in memory output "Q2" at address location "0".
(n) Write a "low" in memory output "Q2" at address location "0".
(o) Write a "high" in memory output "Q3" at address location "0".
(p) Read a "low" in all other memory bit locations.
(q) Read a "high" in memory output "Q3" at address location "0".
(r) Write a "low" in memory output "Q3" at address location "0".
(s) Write a "high" in memory output "Q0" at address location "1".
(t) Read a "low" in all other memory bit locations.
(u) Continue the pattern until a "high" has been walked through every memory bit location.

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3/ A program that fulfills the walking-zero test pattern would be as follows

(a) Write "highs" in all memory address locations.
(b) Read "highs" in all memory address locations.
(c) Write a "low" in memory output "Q0" at address location "0".
(d) Read "high" in all other memory bit locations.
(e) Read a "low" in memory output "Q0" at address location "0".
(f) Write a "high" in memory output "Q0" at address location "0".
(g) Read a "high" in all other memory bit locations.
(h) Write a "low" in memory output "Q1" at address location "0".
(i) Read a "high" in all other memory bit locations.
(j) Read a "low" in memory output "Q1" at address location "0".
(k) Write a "high" in memory output "Q1" at address location "0".
(l) Write a "low" in memory output "Q2" at address location "0".
(m) Read a "high" in all other memory bit locations.
(n) Read a "low" in memory output "Q2" at address location "0".
(o) Write a "high" in memory output "Q2" at address location "0".
(p) Write a "low" in memory output "Q3" at address location "0".
(q) Read a "high" in all other memory bit locations.
(r) Read a "low" in memory output "Q3" at address location "0".
(s) Write a "high" in memory output "Q3" at address location "0".
(t) Write a "low" in memory output "Q0" at address location "1".
(u) Read a "high" in all other memory bit locations.
(v) Continue the pattern until a "low" has been walked through every memory bit location.

4/ For test conditions R1 and R2, load output pins with the load shown on figure 12, note 4.

5/ For test conditions R3 and R4, load output pins with the load shown on figure 12, note 5.

6/ For test conditions R1 and R3, set V_{CC} = 4.5 V ±1 percent.

7/ For test conditions R2 and R4, set V_{CC} = 5.5 V ±1 percent.

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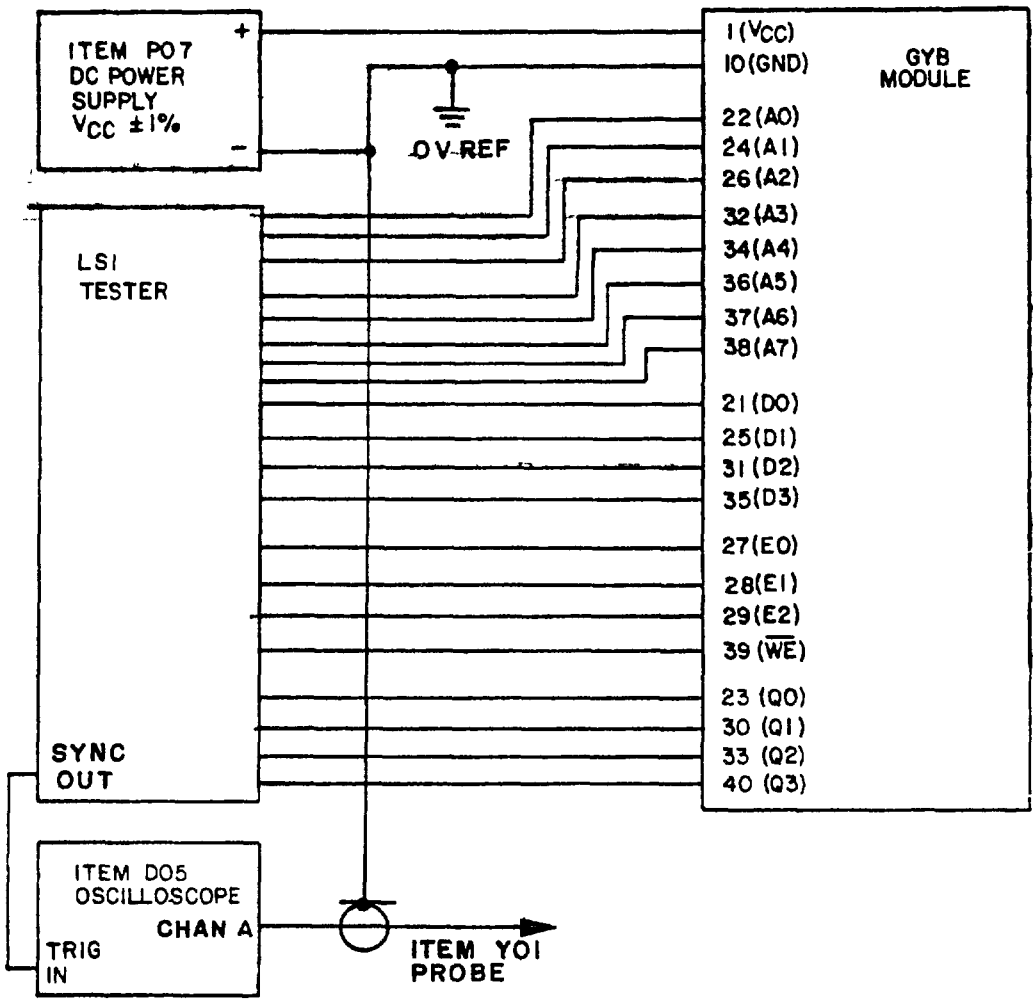


FIGURE 20. Walking-one-walking-zero test pattern test circuit for table XV.

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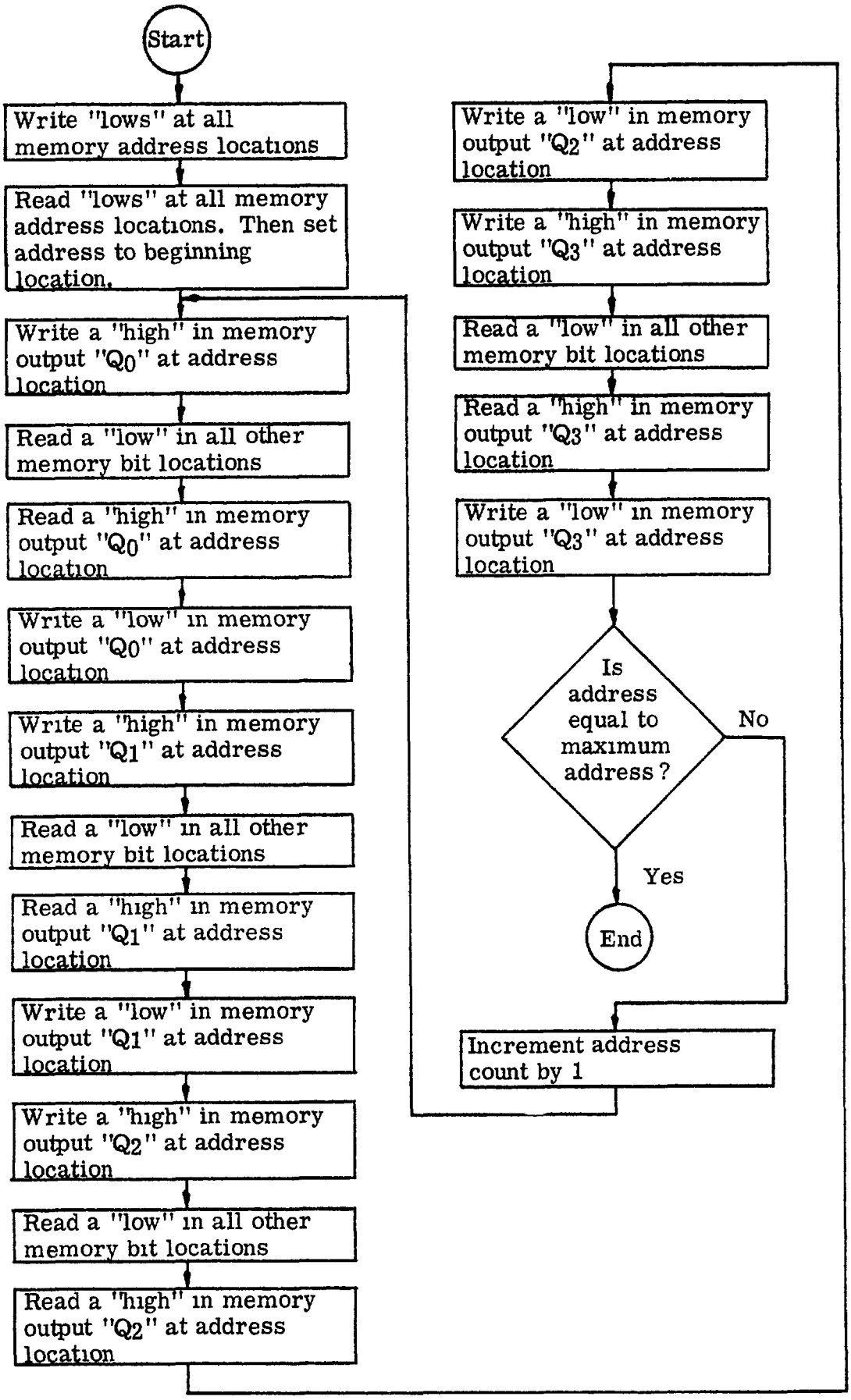


FIGURE 21. Walking-one test pattern flow chart.

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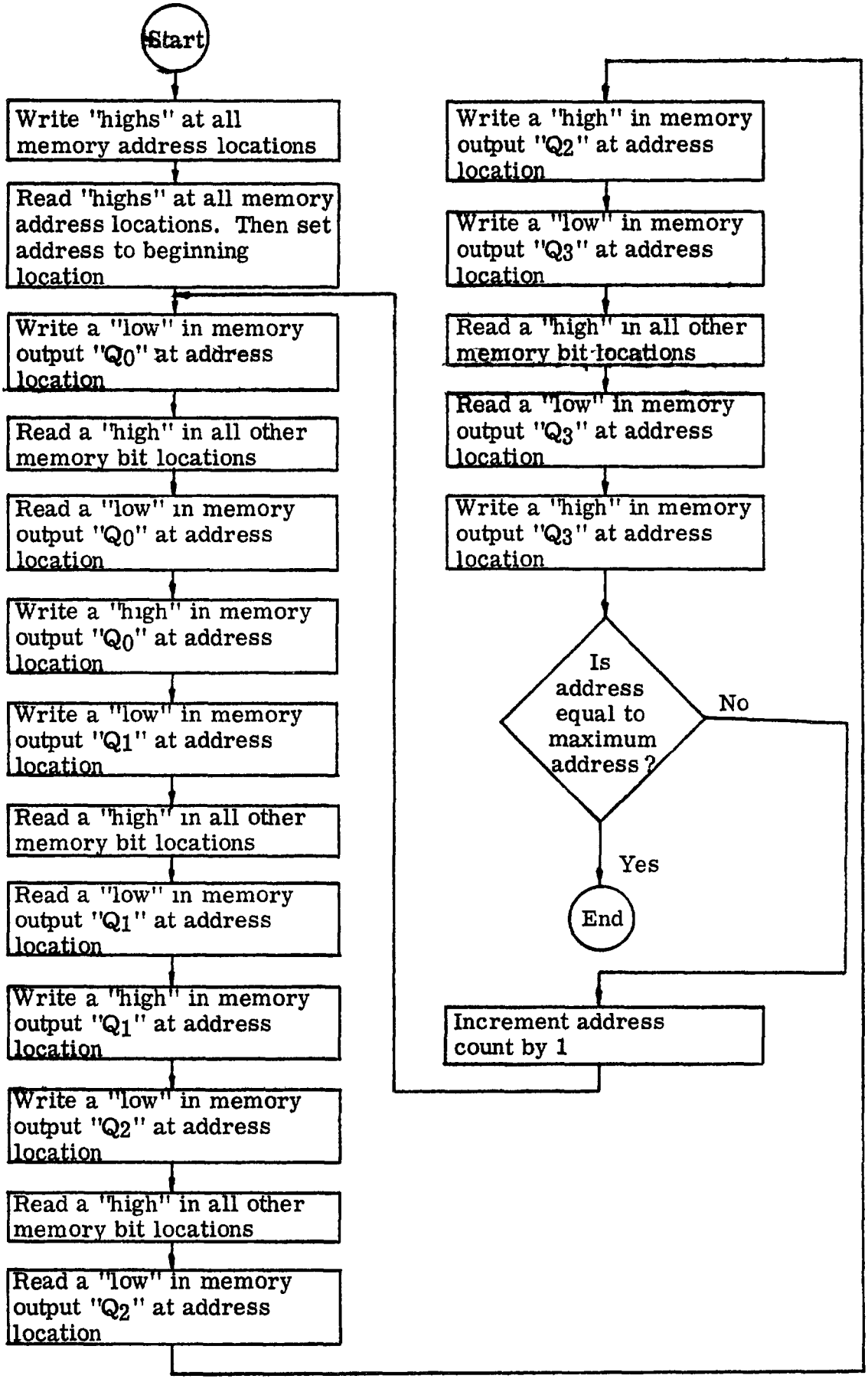


FIGURE 22. Walking-zero test pattern flow chart.

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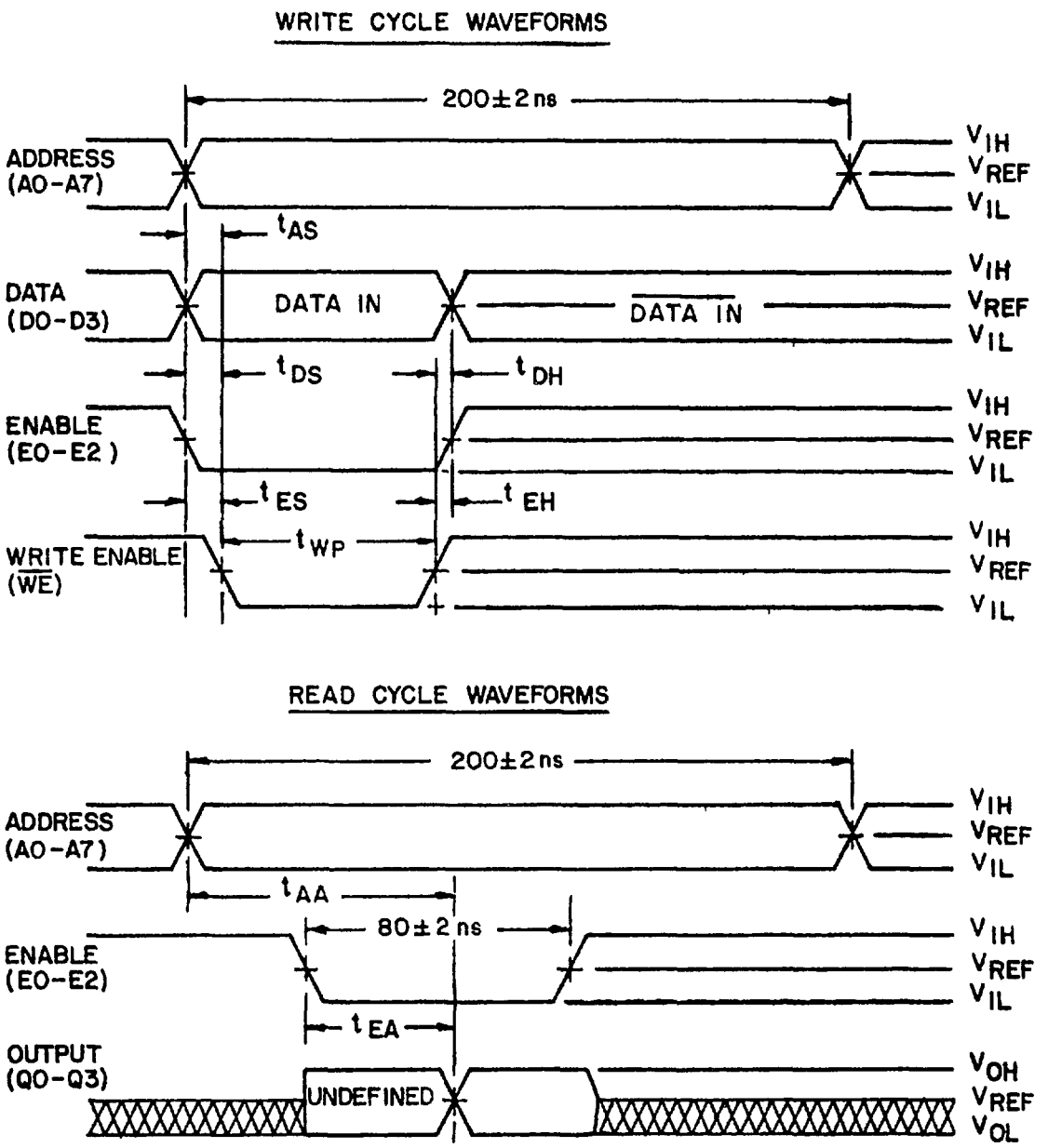


FIGURE 23. Walking-one-walking-zero test pattern test circuit waveforms for table XV.

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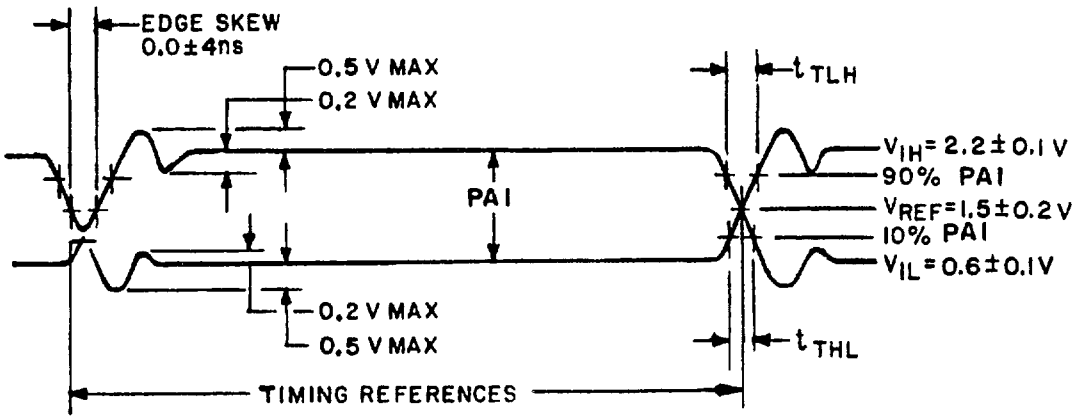
NOTES:

1.	Address setup time	(t_{AS})	= 15 +0, -10 ns
2.	Data setup time	(t_{DS})	= 15 +0, -10 ns
3.	Enable setup time	(t_{ES})	= 15 +0, -10 ns
4.	Write pulse width	(t_{WP})	= 65 ±3 ns
5.	Data hold time	(t_{DH})	= 10 +0, -10 ns
6.	Enable hold time	(t_{EH})	= 10 +0, -10 ns
7.	Address access time (50 pF)	(t_{AA})	= 70 +0, -10 ns
8.	Enable access time (50 pF)	(t_{EA})	= 45 +0, -10 ns
9.	Address access time (150 pF)	(t_{AA})	= 80 +0, -10 ns
10.	Enable access time (150 pF)	(t_{EA})	= 55 +0, -10 ns
11.	High-impedance state		≈ 1.6 V

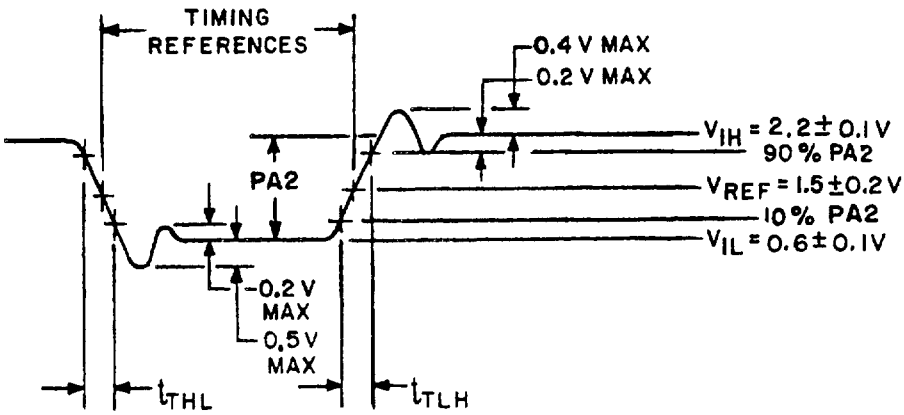
12. Dynamic test input waveforms shall be as specified on figure 24.
13. All address lines need not change simultaneously each cycle.
14. All times must be set per 4.4.3.

FIGURE 23. Walking-one-walking-zero test pattern test circuit waveforms for table XV -Continued.

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BIDIRECTIONAL INPUT WAVEFORMS



UNIDIRECTIONAL INPUT WAVEFORMS

NOTES:
1. t_{THL} = t_{TLH} = 10 ns ±4 ns.
2. All timing references are made at the V_{REF} level.

*FIGURE 24. Dynamic test input waveform requirements.

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TABLE XVI. Galloping-one-galloping-zero test pattern test.

Test condition	Parameter		Initial limits 25°C ±5°C ambient		Test figure
			Min	Max	
S1	Galloping-one-galloping-zero test pattern (C_L = 50 pF, V_{CC} = 4.5 V)	1/, 2/, 3/, 4/, 6/	1/ - 7/		25,26 27,28
S2	Galloping-one-galloping-zero test pattern (C_L = 50 pF, V_{CC} = 5.5 V)	1/, 2/, 3/, 4/, 7/	"		"
S3	Galloping-one-galloping-zero test pattern (C_L = 150 pF, V_{CC} = 4.5 V)	1/, 2/, 3/, 5/, 6/	"		"
S4	Galloping-one-galloping-zero test pattern (C_L = 150 pF, V_{CC} = 5.5 V)	1/, 2/, 3/, 5/, 7/	"		"

1/ Galloping-one-galloping-zero pattern testing. The objective of galloping-one-galloping-zero pattern testing is to test access time. It is necessary to check all possible address transitions (addressing) from a read cycle to a read cycle. This pattern also verifies minimum write pulse width (t_{WP}), address setup time (t_{AS}), data setup time (t_{DS}), enable setup time (t_{ES}), data hold time (t_{DH}), enable hold time (t_{EH}), enable access time (t_{EA}), and address access time (t_{AA}).

2/ A program that fulfills the galloping-one test pattern would be as follows:

(a) Write "lows" at all memory address locations.
(b) Read "lows" at all memory address locations.
(c) Write "highs" at memory address location "0".
(d) Read "lows" at memory address location "1".
(e) Read "highs" at memory address location "0".
(f) Read "lows" at memory address location "2".
(g) Read "highs" at memory address location "0".
(h) Read "lows" at memory address location "3".
(i) Read "highs" at memory address location "0".
(j) Continue the pattern until memory address location "0" has been checked with respect to all other memory address locations.
(k) Write "lows" at memory address location "0".
(l) Write "highs" at memory address location "1".
(m) Read "lows" at memory address location "2".
(n) Read "highs" at memory address location "1".
.
.
.
(o) Read "lows" at memory address location "0".
(p) Read "highs" at memory address location "1".
(q) Write "lows" at memory address location "1".
(r) Write "highs" at memory address location "2".
(s) Continue the pattern until every memory address location has been checked with respect to all other memory address locations.

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3/ A program that fulfills the galloping-zero test pattern would be as follows:

(a) Write "highs" at all memory address locations.
(b) Read "highs" at all memory address locations.
(c) Write "lows" at memory address location "0".
(d) Read "highs" at memory address location "1".
(e) Read "lows" at memory address location "0".
(f) Read "highs" at memory address location "2".
(g) Read "lows" at memory address location "0".
(h) Read "highs" at memory address location "3".
(i) Read "lows" at memory address location "0".
(j) Continue the pattern until memory address location "0" has been checked with respect to all other memory address locations.
(k) Write "highs" at memory address location "0".
(l) Write "lows" at memory address location "1".
(m) Read "highs" at memory address location "2".
(n) Read "lows" at memory address location "1".
. . .
(o) Read "highs" at memory address location "0".
(p) Read "lows" at memory address location "1".
(q) Write "highs" at memory address location "1".
(r) Write "lows" at memory address location "2".
(s) Continue the pattern until every memory address location has been checked with respect to all other memory address locations.

4/ For test conditions S1 and S2, load output pins with the load shown on figure 12, note 4.

5/ For test conditions S3 and S4, load output pins with the load shown on figure 12, note 5.

6/ For test conditions S1 and S3, set V_{CC} = 4.5 V ± percent.

7/ For test conditions S2 and S4, set V_{CC} = 5.5 V ± percent.

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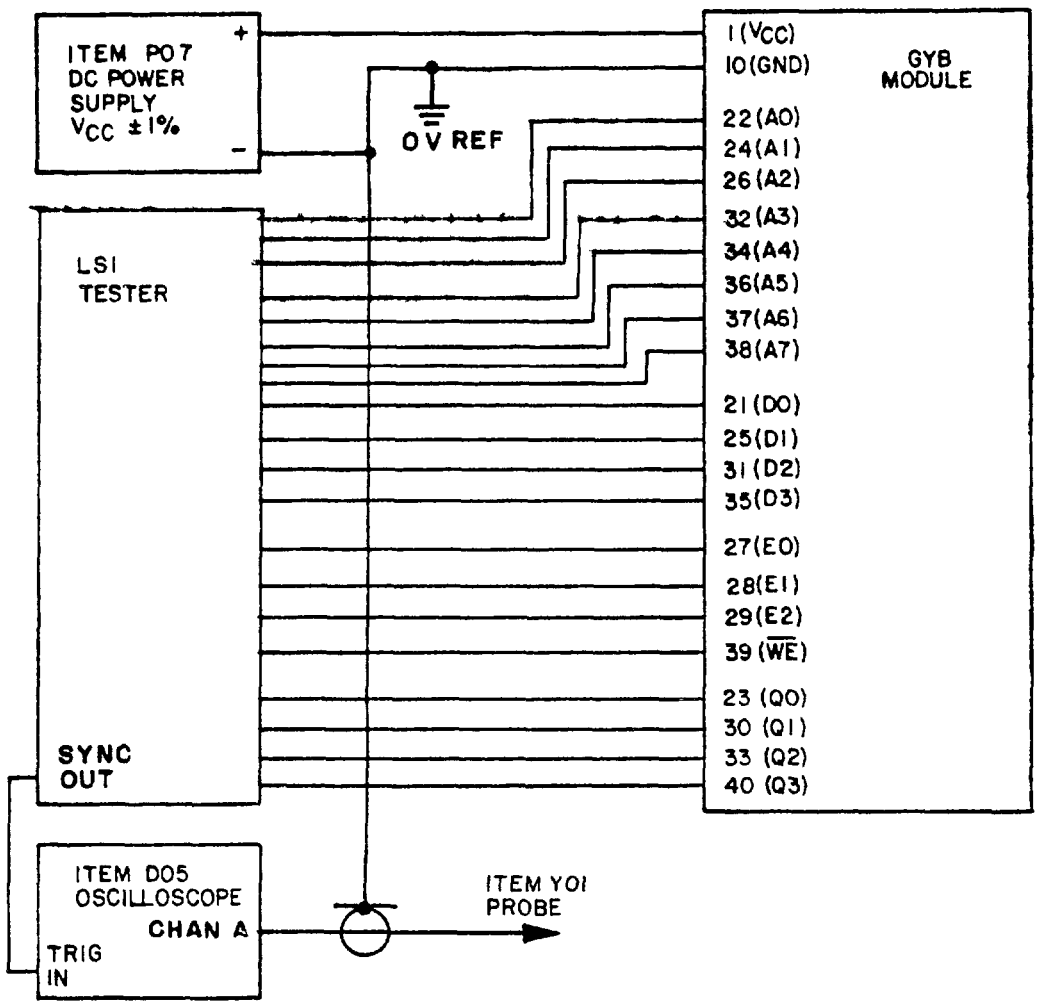


FIGURE 25. Galloping-one-galloping-zero test pattern test circuit for table XVI.

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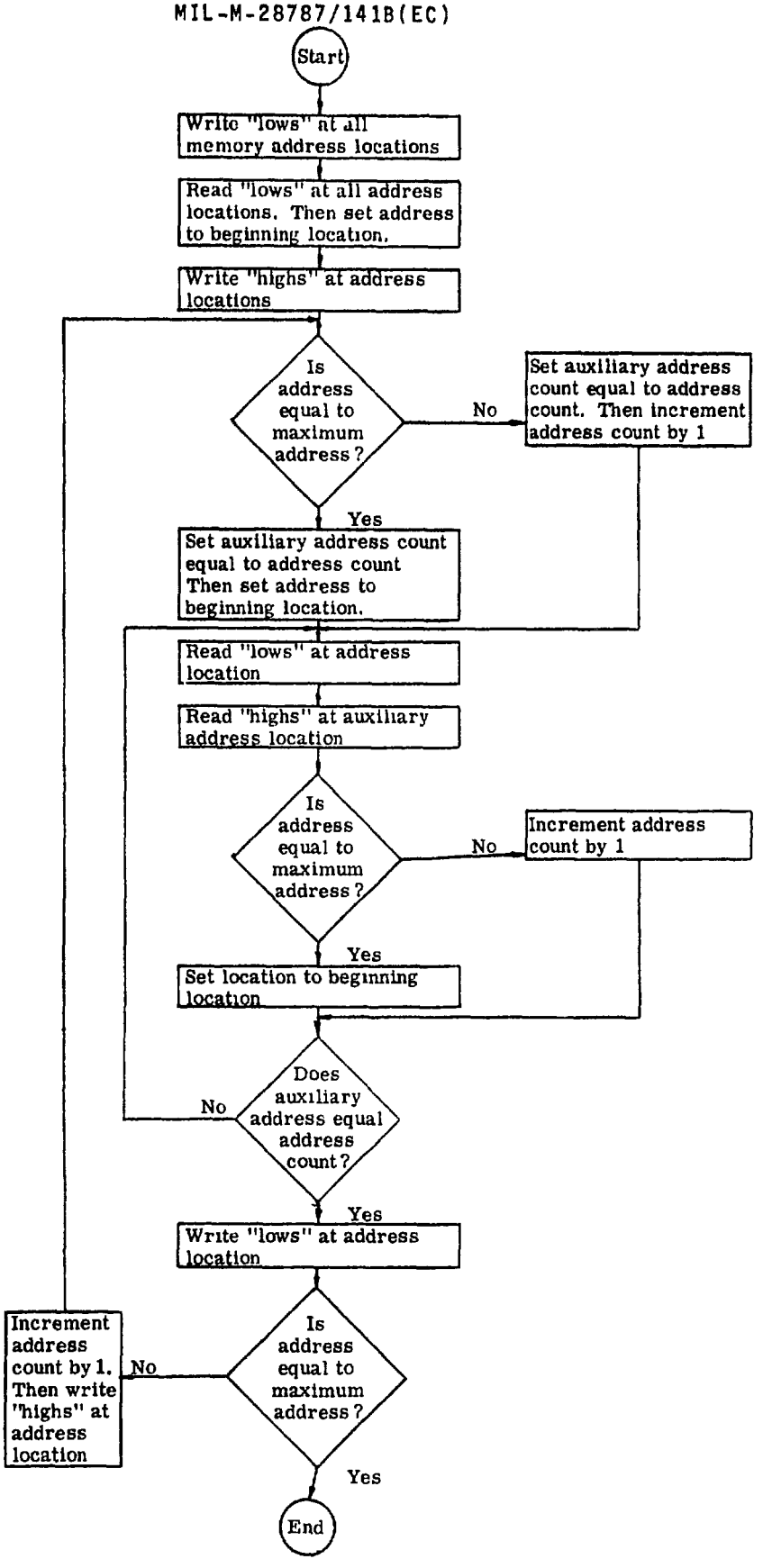


FIGURE 26. Galloping-one pattern flow chart.

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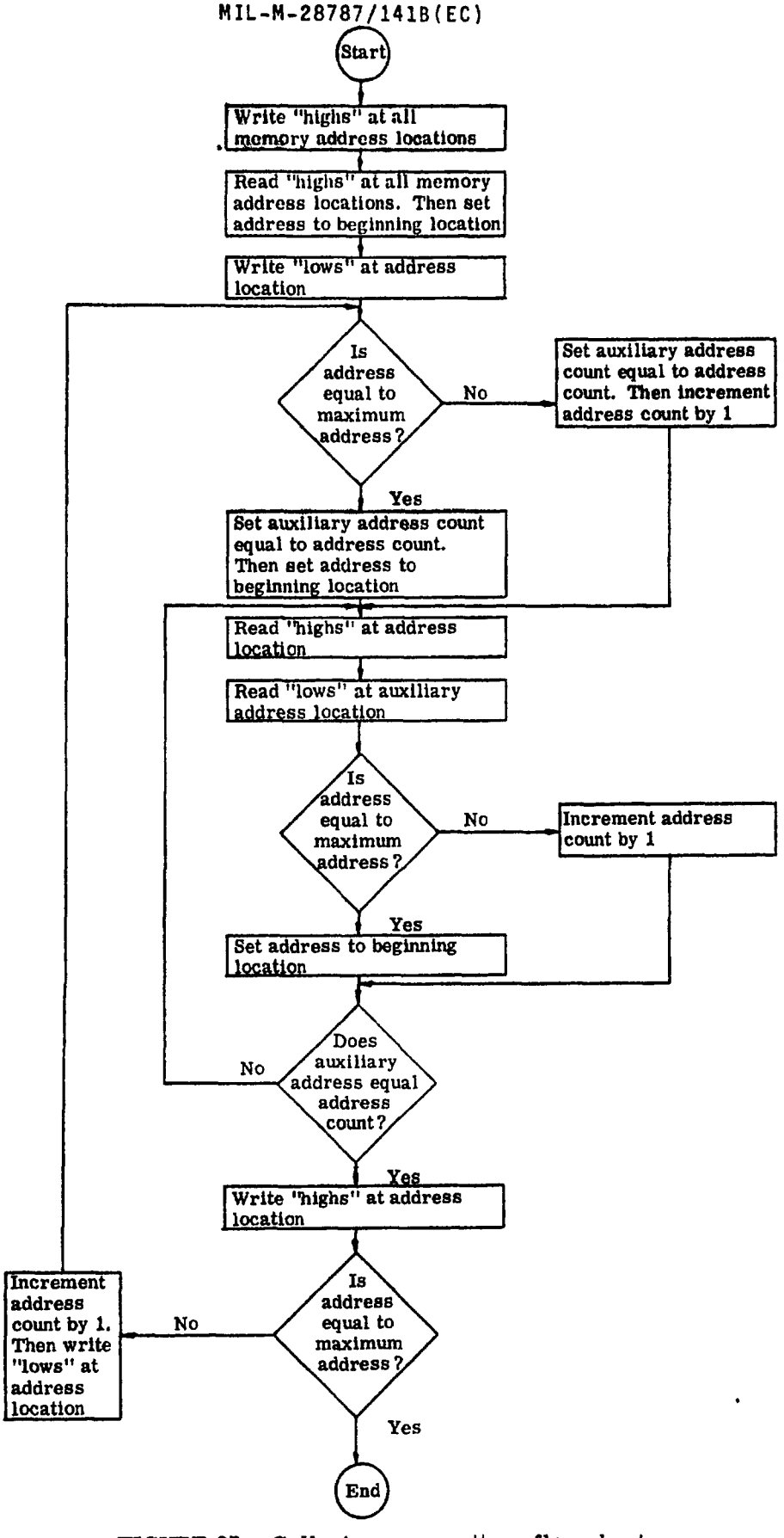


FIGURE 27. Galloping-zero pattern flow chart.

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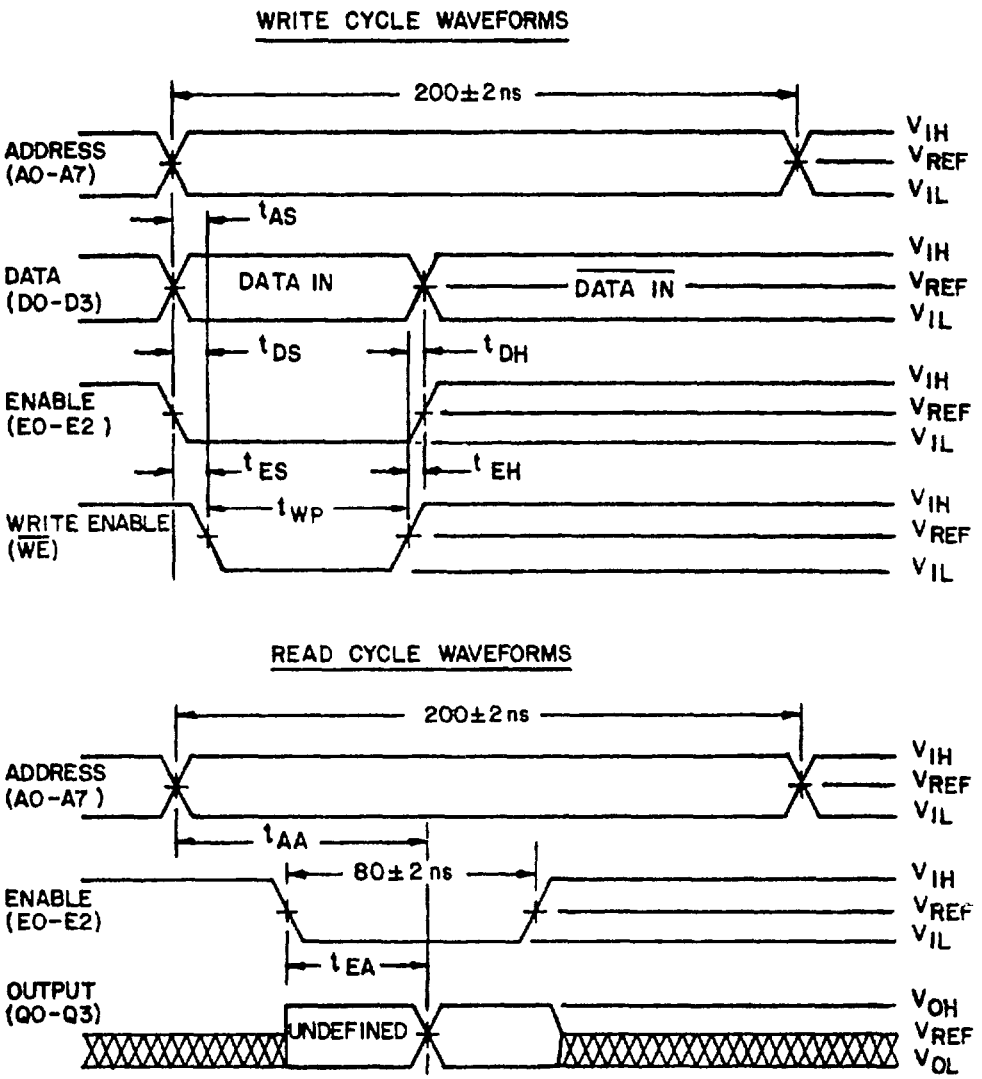


FIGURE 28. Galloping-one-galloping-zero test pattern test circuit waveforms for table XVI.

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NOTES:

1.	Address setup time	(t_{AS})	=	15 +0, -10 ns
2.	Data setup time	(t_{DS})	=	15 +0, -10 ns
3.	Enable setup time	(t_{ES})	=	15 +0, -10 ns
4.	Write pulse width	(t_{WP})	=	65 ±3 ns
5.	Enable hold time	(t_{EH})	=	10 +0, -10 ns
6.	Data hold time	(t_{DH})	=	10 +0, -10 ns
7.	Address access time (50 pF)	(t_{AA})	=	70 +0, -10 ns
8.	Enable access time (50 pF)	(t_{EA})	=	45 +0, -10 ns
9.	Address access time (150 pF)	(t_{AA})	=	80 +0, -10 ns
10.	Enable access time (150 pF)	(t_{EA})	=	55 +0, -10 ns
11.	High-impedance state		≈	1.6 V

12. Dynamic test input waveforms shall be as specified on figure 24.
13. All address lines need not change simultaneously each cycle.
14. All times must be set per 4.4.3.

FIGURE 28. Galloping-one-galloping-zero test pattern test circuit waveforms for table XVI - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:16Z
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*TABLE XVII. Inter-address write-recovery-one-write-recovery-zero test pattern test.

Test condition	Parameter		Initial limits 25°C ±5°C ambient		Test figure
			Min	Max	
T1	Inter-address write-recovery-one-write-recovery-zero test pattern (C_L = 50 pF, V_{CC} = 4.5 V)	1/, 2/, 3/, 4/, 6/	1/ - 7/		29,30 31,32
T2	Inter-address write-recovery-one-write-recovery-zero test pattern (C_L = 50 pF, V_{CC} = 5.5 V)	1/, 2/, 3/, 5/, 6/	"		"
T3	Inter-address write-recovery-one-write-recovery-zero test pattern (C_L = 150 pF, V_{CC} = 4.5 V)	1/, 2/, 3/, 4/, 7/	"		"
T4	Inter-address write-recovery-one-write-recovery-zero test pattern (C_L = 150 pF, V_{CC} = 5.5 V)	1/, 2/, 3/, 5/, 7/	"		"

1/ Inter-address write-recovery-one-write-recovery-zero pattern testing. The objective of write-recovery testing is to check recovery time from a write at one address location to a read at a different address location. The pattern checks recovery time for all possible address transitions from a write cycle to a read cycle. This pattern also verifies data to write enable setup time (t_{DS}), data to write enable hold time (t_{DH}), address to write enable hold time (t_{AH}), enable to write enable hold time (t_{EH}), enable to write enable setup time (t_{ES}), minimum write pulse width (t_{WP}), address access time (t_{AA}), and enable access time (t_{EA}).

2/ A program that fulfills the inter-address write-recovery-zero test pattern would be as follows:

(a) Write "highs" at all memory address locations.
(b) Read "highs" at all memory address locations.
(c) Write "lows" at memory address location "1".
(d) Read "highs" at memory address location "0".
(e) Read "lows" at memory address location "1".
(f) Write "highs" at memory address location "1".
(g) Read "highs" at memory address location "0".
(h) Read "highs" at memory address location "1".
(i) Continue this pattern until memory address "0" has been checked with respect to all other memory address locations.
(j) Write "lows" at memory address location "2".
(k) Read "highs" at memory address location "1".
(l) Read "lows" at memory address location "2".
(m) Write "highs" at memory address location "2".
(n) Read "highs" at memory address location "1".
(o) Read "highs" at memory address location "2".

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2/ Continued.

.
.
.
(p) Write "lows" at memory address location "0".
(q) Read "highs" at memory address location "1".
(r) Read "lows" at memory address locations "0".
(s) Write "highs" at memory address location "0".
(t) Read "highs" at memory address location "1".
(u) Read "highs" at memory address location "0".
(v) Continue the pattern until all memory address locations have been checked with respect to all other memory address locations.

3/ A program that fulfills the inter-address write-recovery-one-test pattern would be as follows:

(a) Write "lows" at all memory address locations.
(b) Read "lows" at all memory address locations.
(c) Write "highs" at memory address location "1".
(d) Read "lows" at memory address location "0".
(e) Read "highs" at memory address location "1".
(f) Write "lows" at memory address location "1".
(g) Read "lows" at memory address location "0".
(h) Read "lows" at memory address location "1".
(i) Continue the pattern until memory address location "0" has been checked with respect to all other memory address locations.
(j) Write "highs" at memory address location "2".
(k) Read "lows" at memory address location "1".
(l) Read "highs" at memory address location "2".
(m) Write "lows" at memory address location "2".
(n) Read "lows" at memory address location "1".
(o) Read "lows" at memory address location "2".
.
.
.
(p) Write "highs" at memory address location "0".
(q) Read "lows" at memory address location "1".
(r) Read "highs" at memory address location "0".
(s) Write "lows" at memory address location "0".
(t) Read "lows" at memory address location "1".
(u) Read "lows" at memory address location "0".
(v) Continue the pattern until all memory address locations have been checked with respect to all other memory address locations.

4/ For test conditions T1 and T3, set V_{CC} = 4.5 V ±1 percent.

5/ For test conditions T2 and T4, set V_{CC} = 5.5 V ±1 percent.

6/ For test conditions T1 and T2, load output pins with the load shown on figure 12, note 4.

7/ For test conditions T3 and T4, load output pins with the load shown on figure 12, note 5.

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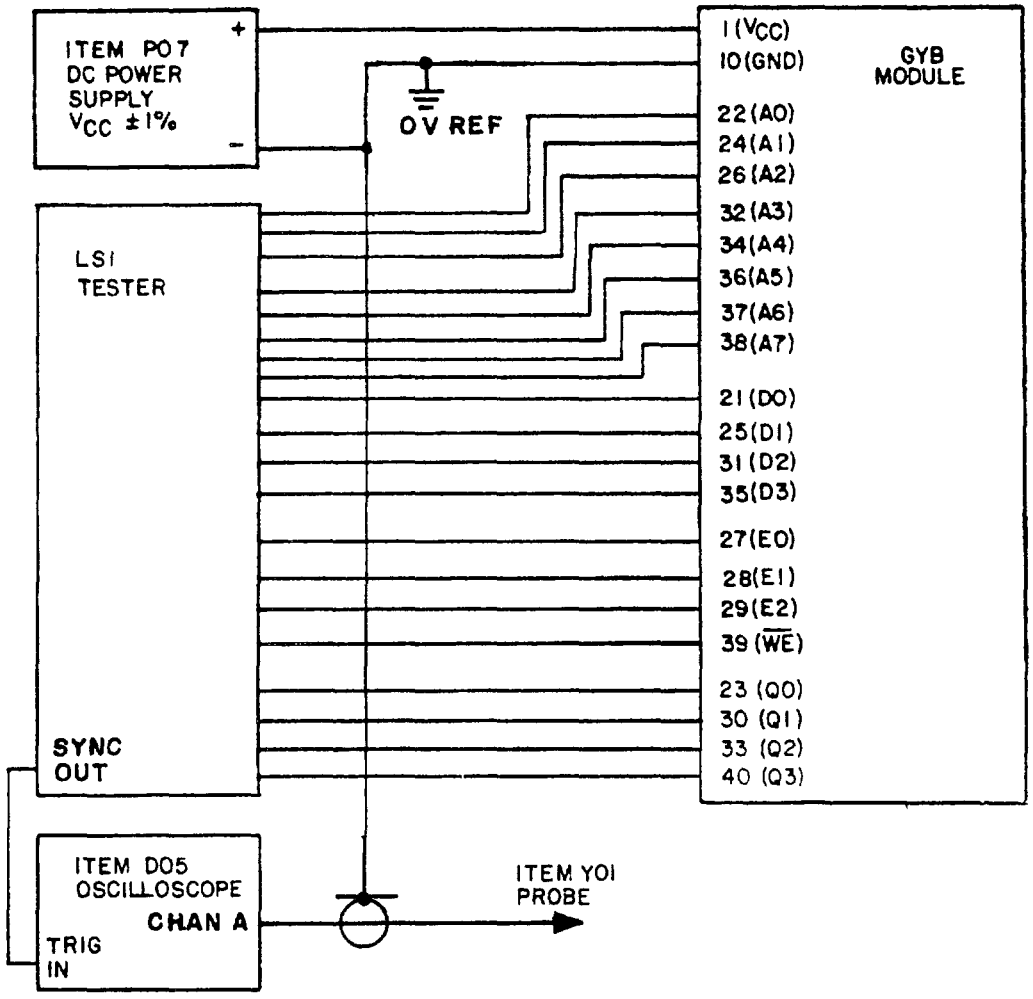


FIGURE 29. Inter-address write-recovery-one-write-recovery-zero test pattern test circuit for table XVII.

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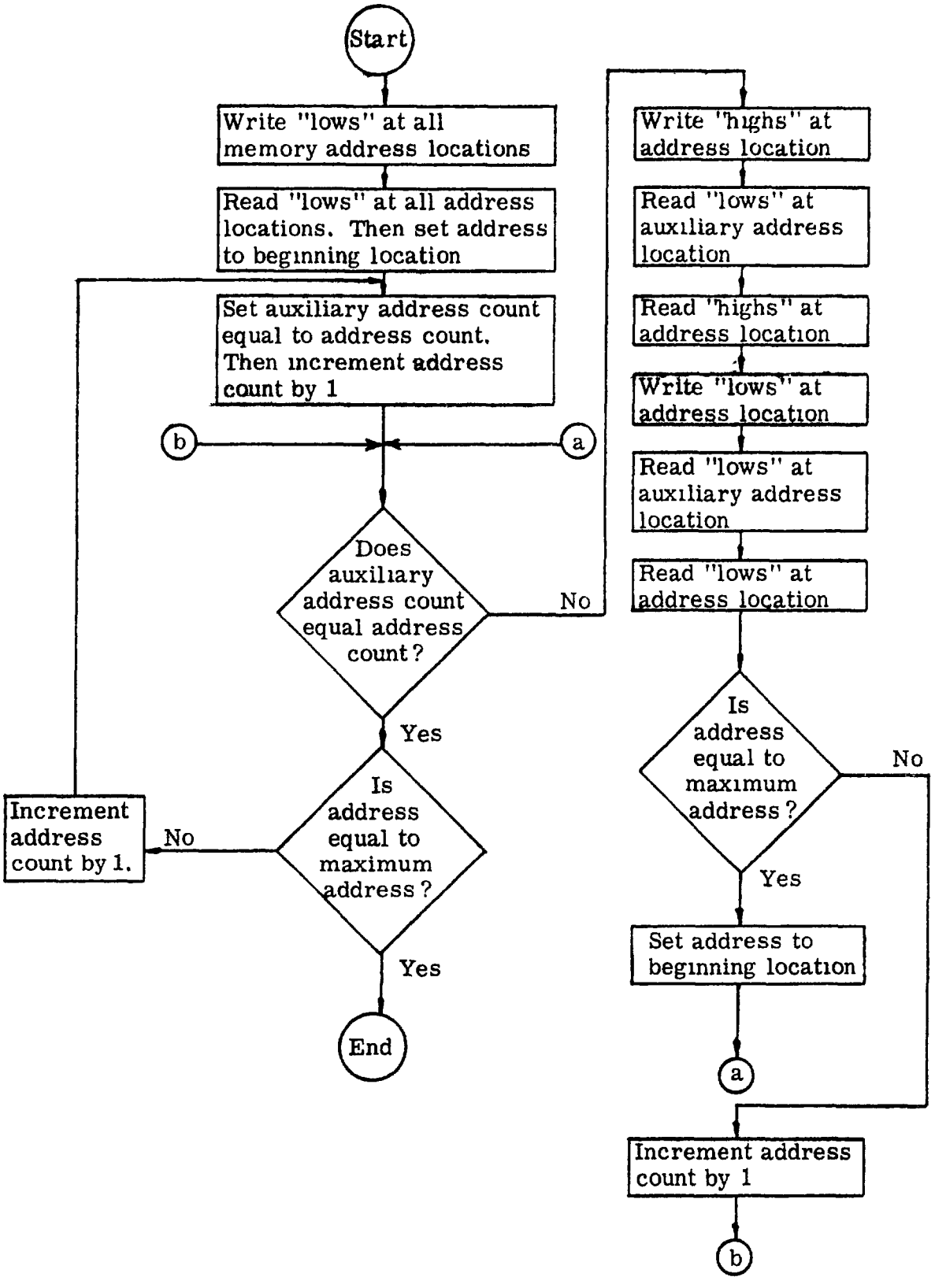


FIGURE 30. Inter-address write-recovery-one pattern flow chart.

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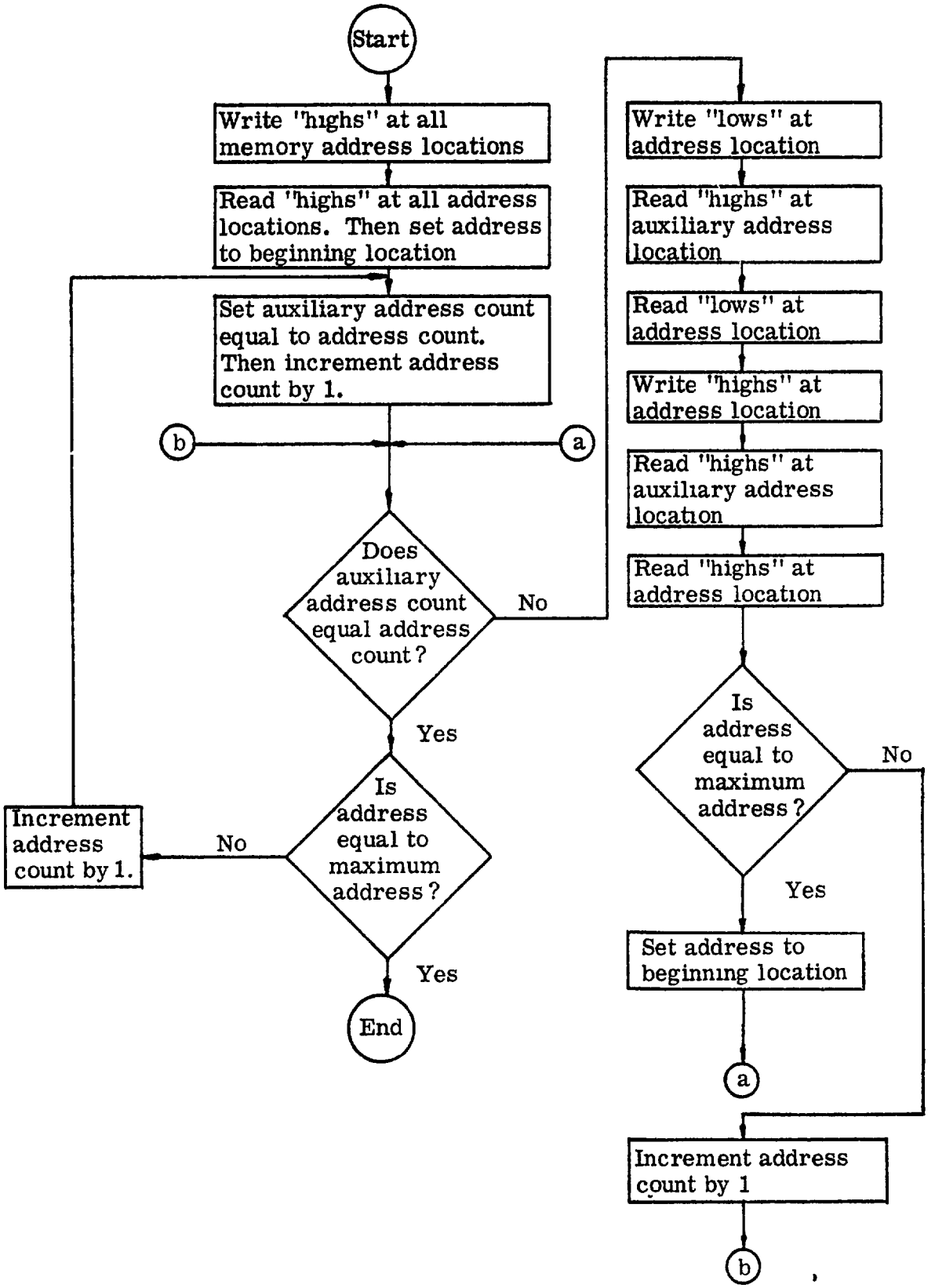


FIGURE 31. Inter-address write-recovery-zero pattern flow chart.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:16Z
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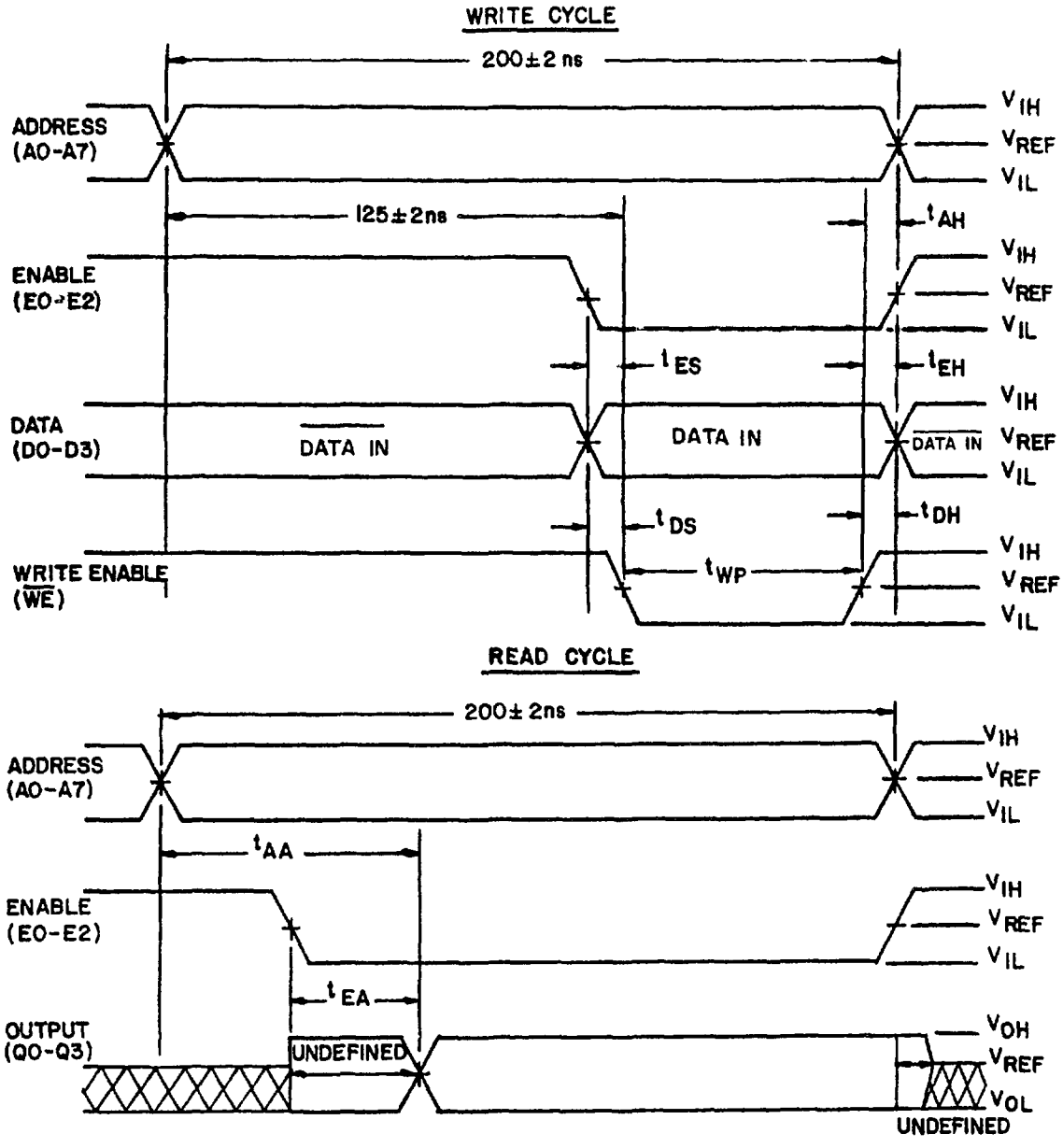


FIGURE 32. Inter-address write-recovery-one-write-recovery-zero test pattern test circuit waveforms for table XVII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:16Z
Check the source to verify that this is the current version before use.

NOTES:

1.	Data setup time	(t_{DS})	=	15 +0, -10 ns
2.	Write pulse width	(t_{WP})	=	65 ±3 ns
3.	Data hold time	(t_{DH})	=	10 +0, -10 ns
4.	Enable hold time	(t_{EH})	=	10 +0, -10 ns
5.	Address hold time	(t_{AH})	=	10 +0, -10 ns
6.	Address access time (50 pF)	(t_{AA})	=	70 +0, -10 ns
7.	Enable access time (50 pF)	(t_{EA})	=	45 +0, -10 ns
8.	Address access time (150 pF)	(t_{AA})	=	80 +0, -10 ns
9.	Enable access time (150 pF)	(t_{EA})	=	55 +0, -10 ns
10.	High-impedance state		≈	1.6 V
11.	Enable setup time	(t_{ES})	=	15 +0, -10 ns

12. Dynamic test input waveforms shall be as specified on figure 24.
13. All address lines need not change simultaneously each cycle.
14. All times must be set per 4.4.3.

FIGURE 32. Inter-address write-recovery-one-write-recovery-zero test pattern test circuit waveforms for table XVII - Continued.

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*TABLE XVIII. Marching write-recovery-one-write-recovery-zero and enable/disable test.

Test condition	Parameter		Initial limits 25°C ±5°C ambient		Test figure
			Min	Max	
U1	Marching write-recovery-one-write-recovery-zero and enable/disable test pattern (C_L = 50 pF, V_{CC} = 4.5 V)	1/, 2/, 3/, 4/, 6/	1/ - 7/		33,34 35,36
U2	Marching write-recovery-one-write-recovery-zero and enable/disable test pattern (C_L = 50 pF, V_{CC} = 5.5 V)	1/, 2/, 3/, 5/, 6/	"		"
U3	Marching write-recovery-one-write-recovery-zero and enable/disable test pattern (C_L = 150 pF, V_{CC} = 4.5 V)	1/, 2/, 3/, 4/, 7/	"		"
U4	Marching write-recovery-one-write-recovery-zero and enable/disable test pattern (C_L = 150 pF, V_{CC} = 5.5 V)	1/, 2/, 3/, 5/, 7/	"		"

1/ Marching write-recovery-one-write-recovery-zero and enable/disable testing. The objective of the marching write-recovery-one-write-recovery-zero and enable/disable testing is to check for the proper functioning of the address and chip select circuitry in a dymanic environment. The pattern checks to assure the module 256 unique word locations that can be independently addressed. The pattern also verifies address setup time (t_{AS}), enable setup time (t_{ES}), enable hold time (t_{EH}), write pulse width (t_{WP}), write recovery time (t_{WR}), and data setup time (t_{DS}).

2/ A program that fulfills the marching write-recovery-one and enable/disable pattern would be as follows:

(a) Write "lows" at all memory address locations.
(b) Read "lows" at all memory address locations.
(c) Read "lows" at address location "0".
(d) Write and read "highs" at address location "0".
(e) Apply disable pulse and attempt to write "lows" at address location "0".
(f) Read "highs" at address location "0".
.
.
.
(g) Read "lows" at address location "255".
(h) Write and read "highs" at address location "255".
(i) Apply disable pulse and attempt to write "lows" at address location "255".
(j) Read "highs" at address location "255".
(k) Read "highs" at address location "255".
(l) Write and read "lows" at address location "255".
(m) Apply disable pulse and attempt to write "highs" at address location "255".
(n) Read "lows" at address location "255".

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:16Z
Check the source to verify that this is the current version before use.

2/ Continued.

.
.
.

(o) Read "highs" at address location "0".
(p) Write and read "lows" at address location "0".
(q) Apply disable pulse and attempt to write "highs" at address location "0".
(r) Read "lows" at address location "0".

3/ A program that fulfills the marching write-recovery-zero and enable/disable pattern would be as follows:

(a) Write "highs" at all memory address locations.
(b) Read "highs" at all memory address locations.
(c) Read "highs" at address location "0".
(d) Write and read "lows" at address location "0".
(e) Apply disable pulse and attempt to write "highs" at address location "0".
(f) Read "lows" at address location "0".

.
.
.

(g) Read "highs" at address location "255".
(h) Write and read "lows" at address location "255".
(i) Apply disable pulse and attempt to write "highs" at address location "255".
(j) Read "lows" at address location "255".
(k) Read "lows" at address location "255".
(l) Write and read "highs" at address location "255".
(m) Apply disable pulse and attempt to write "lows" at address location "255".
(n) Read "highs" at address location "255".

.
.
.

(o) Read "lows" at address location "0".
(p) Write and read "highs" at address location "0".
(q) Apply disable pulse and attempt to write "lows" at address location "0".
(r) Read "highs" at address location "0".

4/ For test conditions U1 and U3, set V_{CC} = 4.5 V ±1 percent.

5/ For test conditions U2 and U4, set V_{CC} = 5.5 V ±1 percent.

6/ For test conditions U1 and U2, load output pins with the load shown on figure 12, note 4.

7/ For test conditions U3 and U4, load output pins with the load shown on figure 12, note 5.

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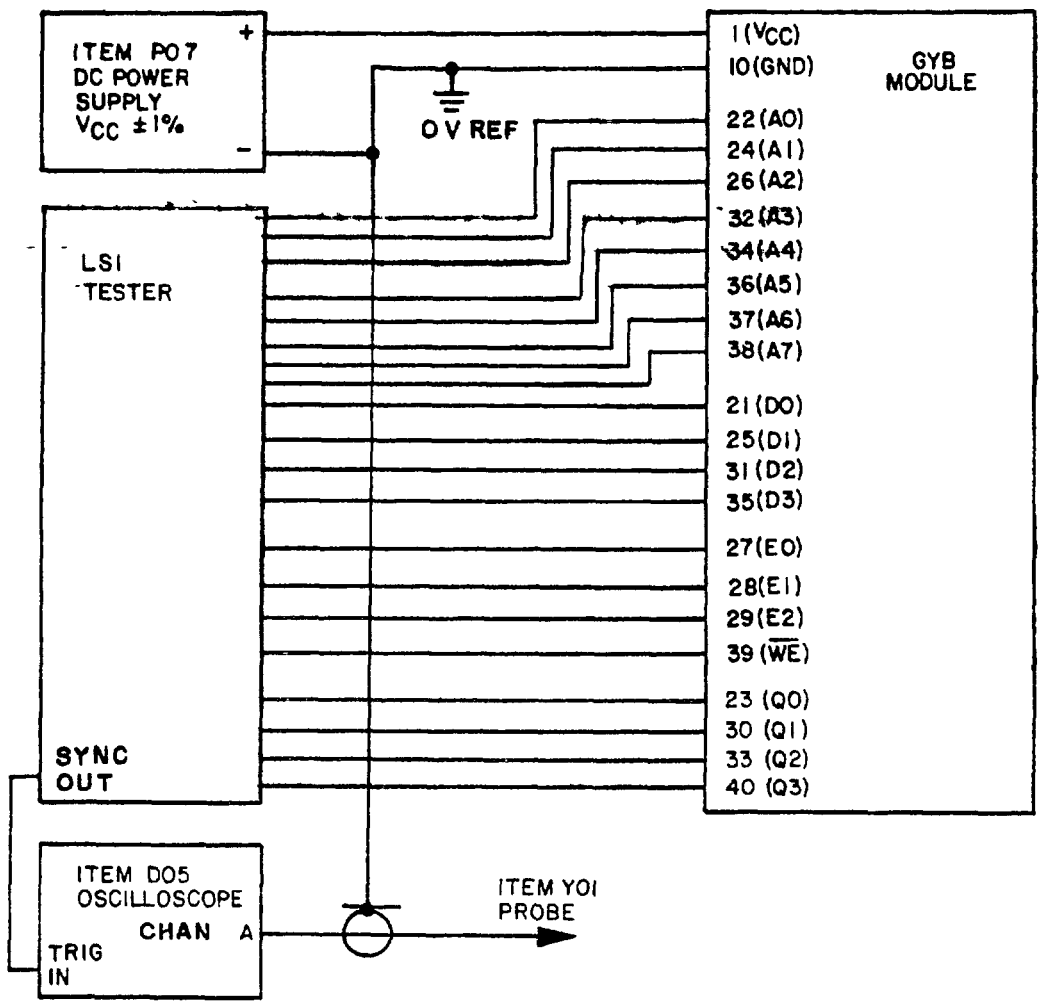


FIGURE 33. Marching write-recovery-one-write-recovery-zero and enable/disable test circuit for table XVIII.

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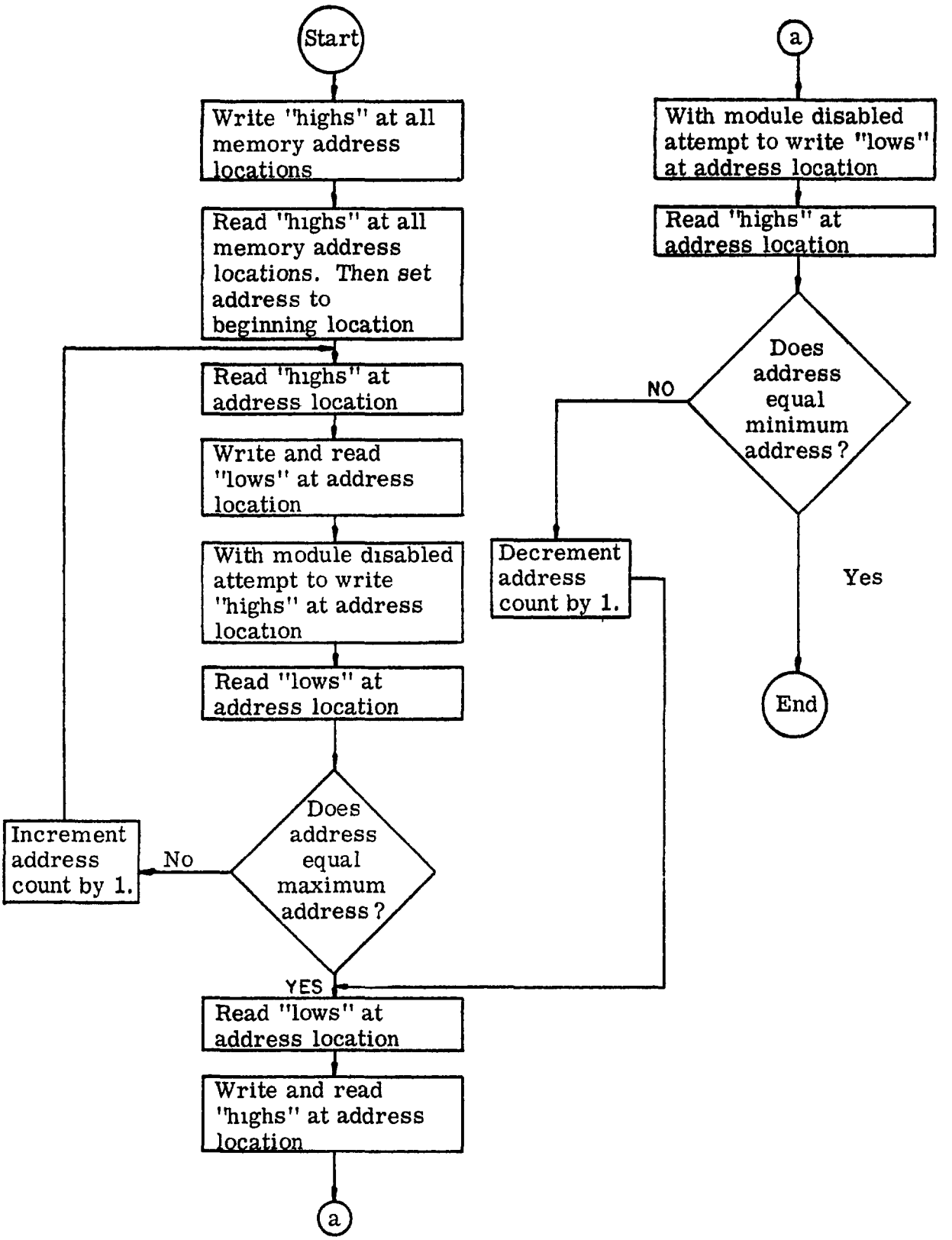


FIGURE 34. Marching write-recovery-zero and enable/disable pattern flow chart.

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Check the source to verify that this is the current version before use.

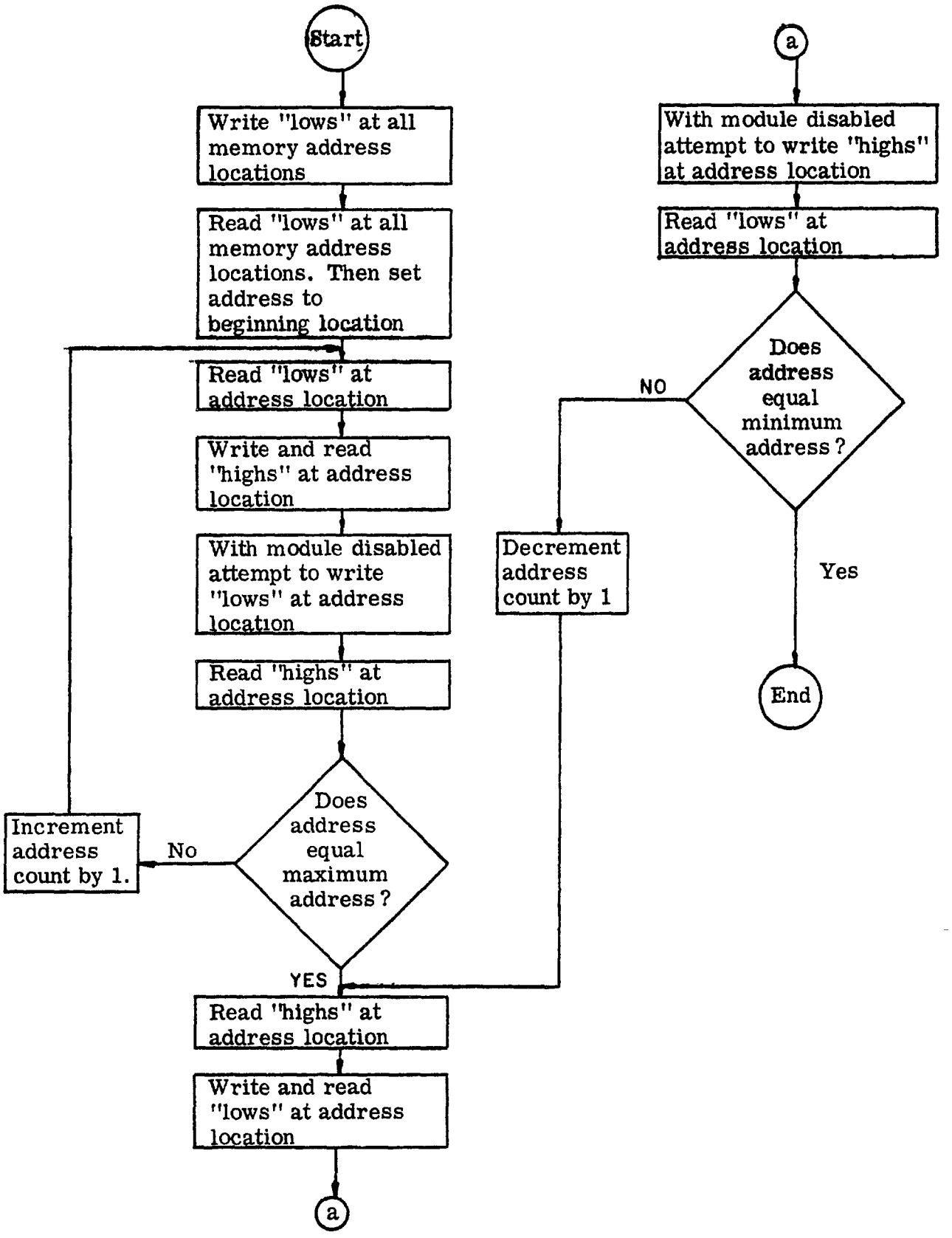


FIGURE 35. Marching write-recovery-one and enable/disable pattern flow chart.

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Check the source to verify that this is the current version before use.

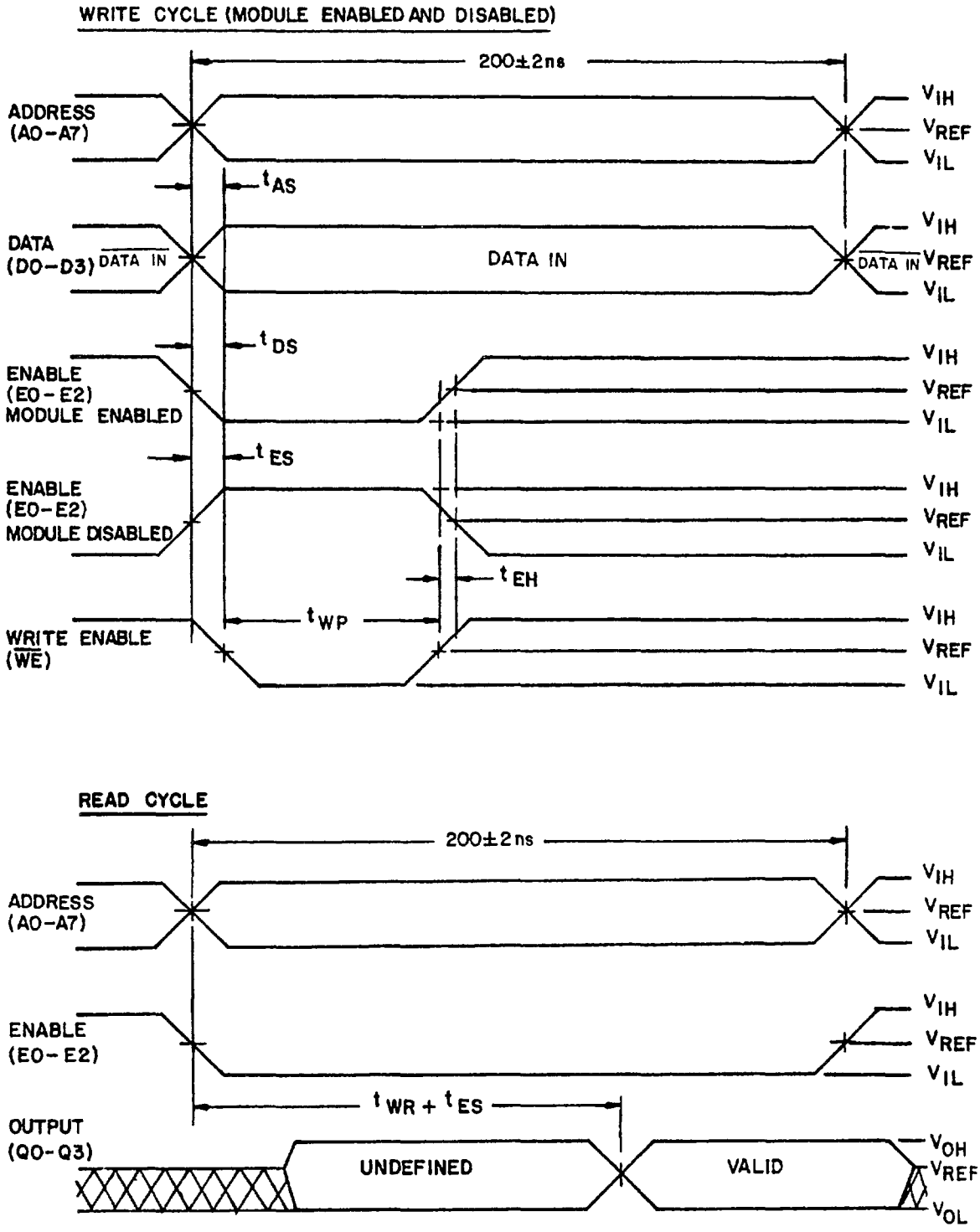
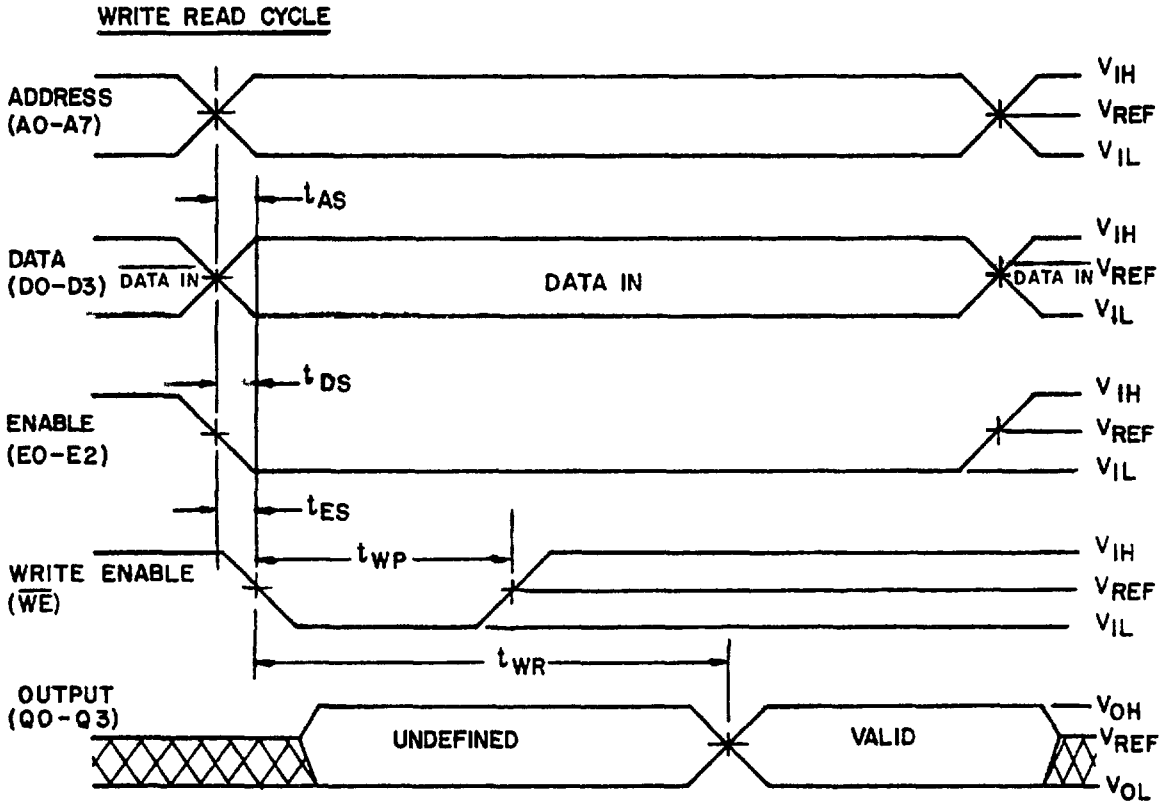


FIGURE 36. Marching write-recovery-one-write-recovery-zero and enable/disable test circuit waveforms for table XVIII.

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NOTES:

1. Data setup time (t_{DS}) = 15 +0, -10 ns
2. Address setup time (t_{AS}) = 15 +0, -10 ns
3. Enable setup time (t_{ES}) = 15 +0, -10 ns
4. Enable hold time (t_{EH}) = 10 +0, -10 ns
5. Write pulse width (t_{WP}) = 65 ±3 ns
6. Write recovery (50 pF) (t_{WR}) = 120 +0, -10 ns
7. Write recovery (150 pF) (t_{WR}) = 130 +0, -10 ns
8. High-impedance state ≈ 1.6 V
9. Dynamic test input waveforms shall be as specified on figure 24.
10. All address lines need not change simultaneously each cycle.
11. All times must be set per 4.4.3.

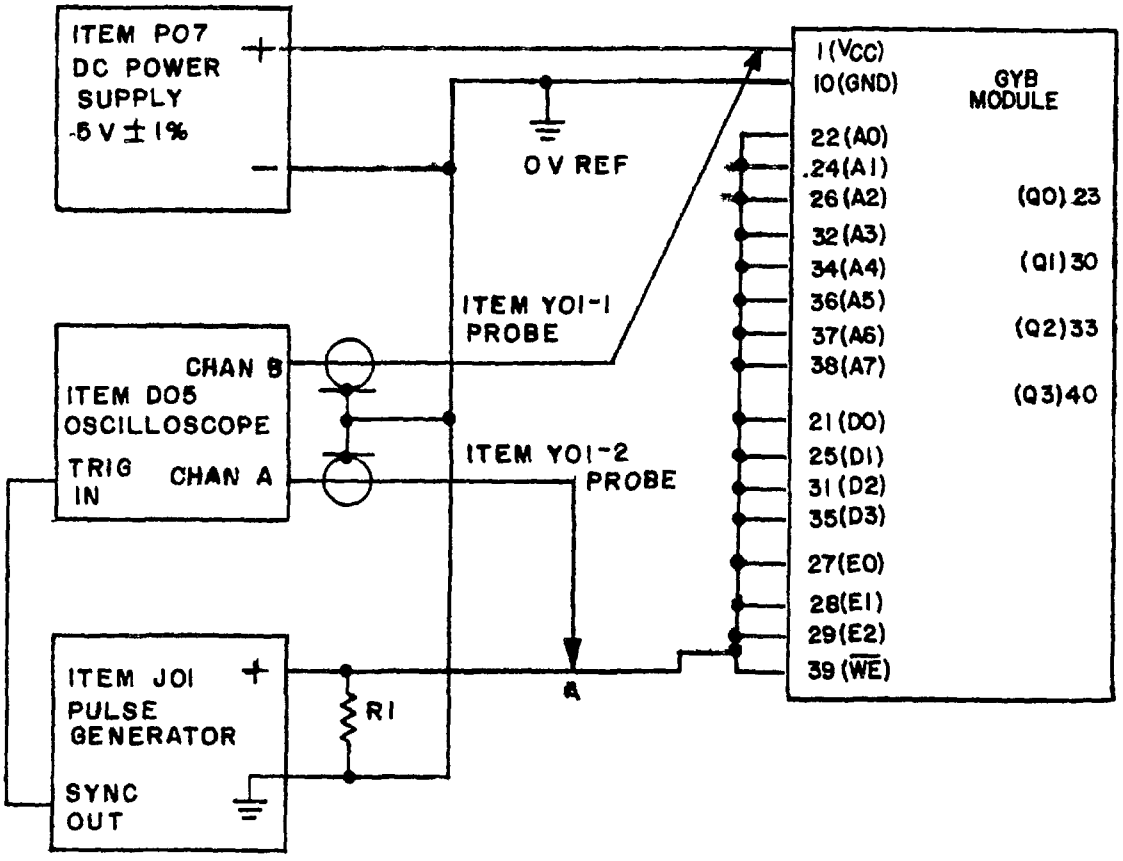
FIGURE 36. Marching write-recovery-one-write-recovery-zero and enable/disable test circuit waveforms for table XVIII -Continued.

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TABLE XIX. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
V	Power supply transient amplitude	0	400	mV	37,38

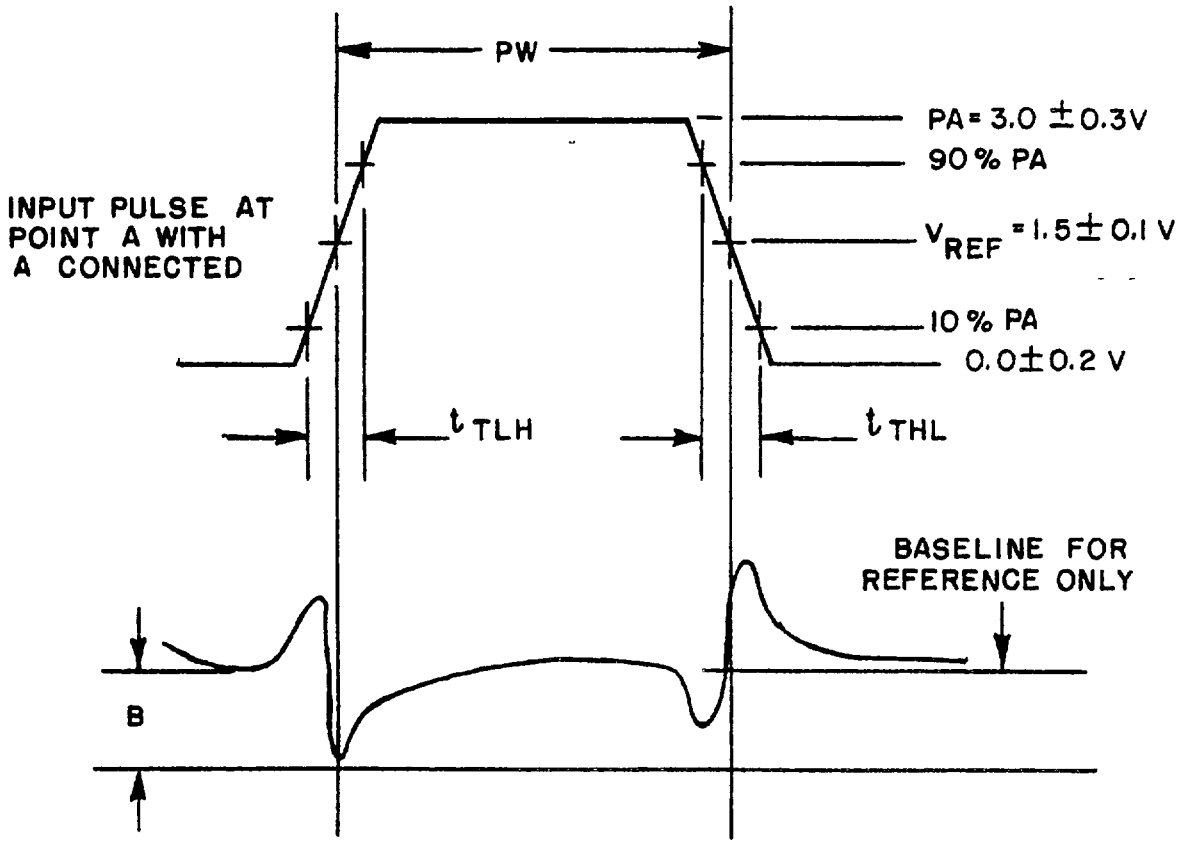
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:16Z
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NOTE: R1 = 51.1Ω ±5%, 1 W.

*FIGURE 37. Power supply transient amplitude test circuit for table XIX.

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NOTES:
1. Input pulse parameters:
   PW = 200 ±20 ns.
   $t_{TLH} = t_{THL}$ = 20 ±2 ns (at point A).
   PRR = 1 MHz ±10%.
2. B = Measurement from the baseline to the most negative excursion of the waveform.
3. All voltage levels are referenced to ground.

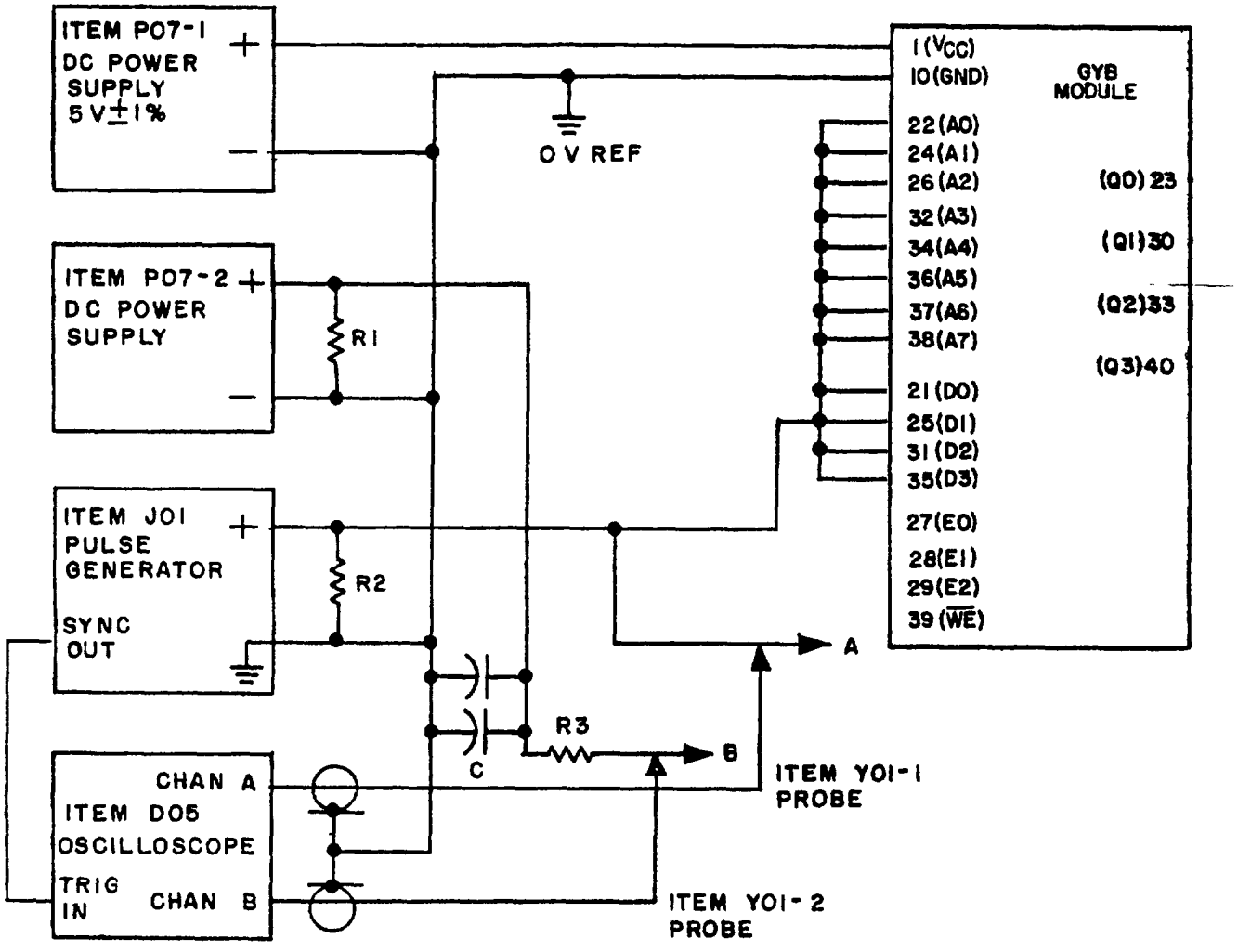
*FIGURE 38. Power supply transient amplitude test circuit waveforms for table XIX.

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TABLE XX. Input crosstalk (low-level and high-level) tests.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
W1	E0 Input crosstalk (low-level)	28,29,39	27	0	700	mV	39,40
W2	E1 "	27,29,39	28	"	"	"	"
W3	E2 "	27,28,39	29	"	"	"	"
W4	$\overline{WE}$ "	27,28,29	39	"	600	"	"
W5	E0 Input crosstalk (high-level)	28,29,39	27	"	"	"	"
W6	E1 "	27,29,39	28	"	"	"	"
W7	E2 "	27,28,39	29	"	"	"	"
W8	$\overline{WE}$ "	27,28,29	39	"	500	"	"

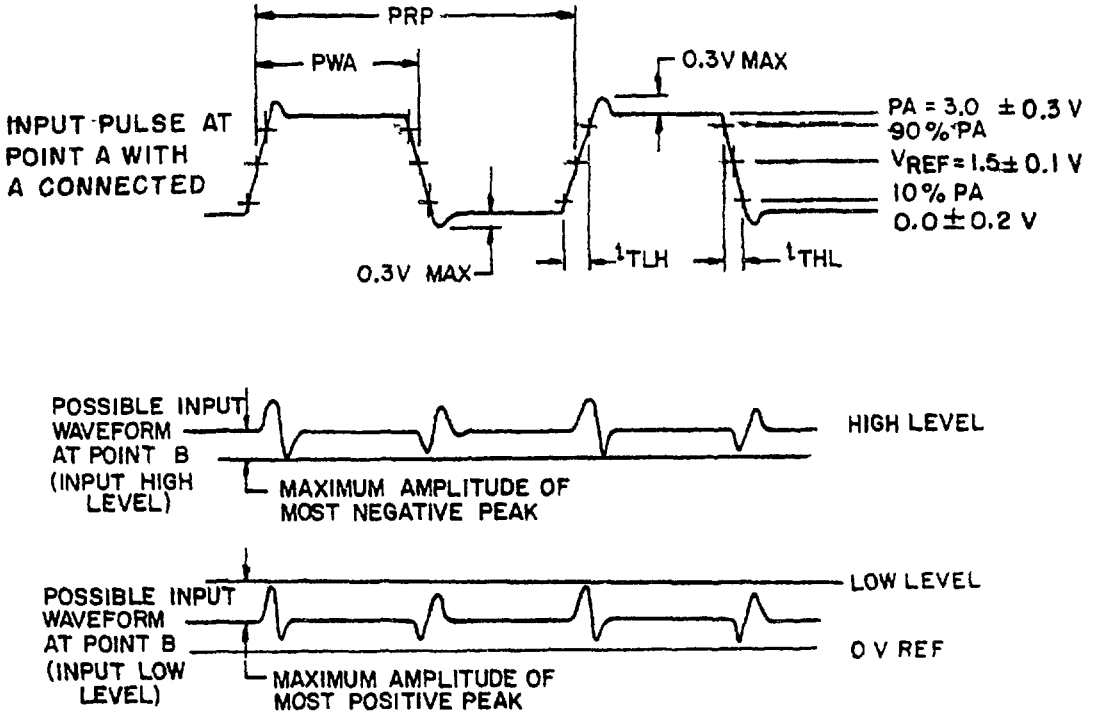
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Check the source to verify that this is the current version before use.



NOTES:
1. R1 = R2 = 51.1Ω ±5%, 1 W.
2. Load all outputs with the load shown on figure 12.
3. R3 = 150Ω ±1%, 1/8 W.
4. Add sufficient capacitance to reduce item P07-2 power supply noise to < 50 mV.
5. For test conditions W1-W4 set item P07-2 to 0.45 V ±1%.
6. For test conditions W5-W8 set item P07-2 to 2.4 V ±1%.

FIGURE 39. Input crosstalk (low-level and high-level) test circuit for table XX.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:16Z
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NOTES:
1. Input pulse parameters:
   $t_{THL} = t_{TLH} = 20 \pm 4$ ns.
   PWA = 500 ±50 ns.
   PRP = 1,000 ±100 ns.
2. The lead length to all input pins and output pins shall be less than 2 inches (50.80 mm).

*FIGURE 40. Input crosstalk (low-level and high-level) test circuit waveforms for table XX.

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TABLE XXI. Power supply filter capacitance test.

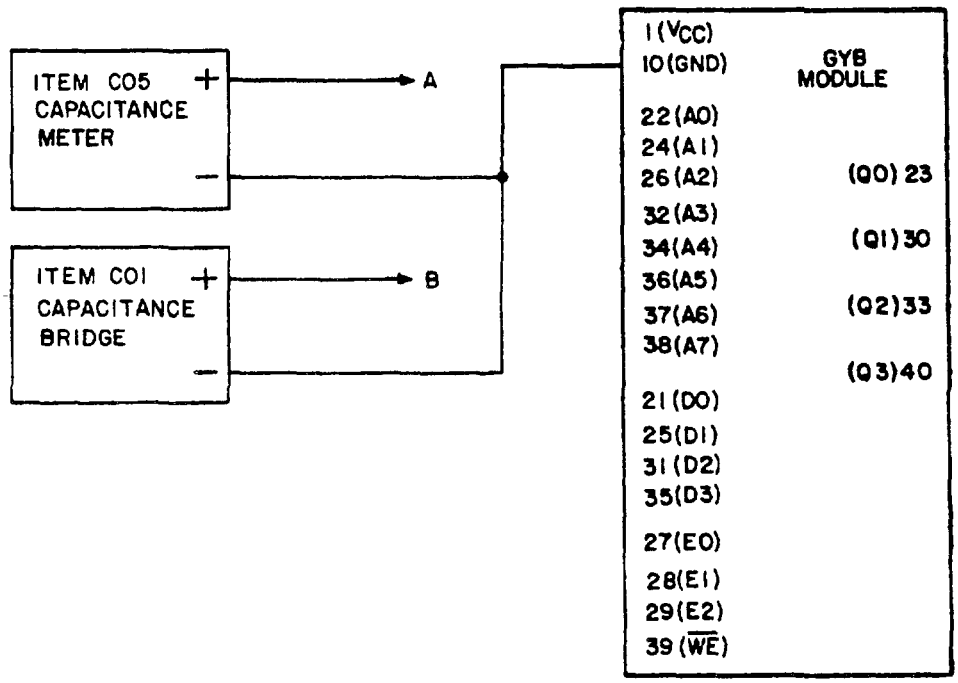
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
X	Power supply filter capacitance	Open	1	1.65	2.75	µF	41

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Check the source to verify that this is the current version before use.

TABLE XXII. Input and output capacitance test.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
Y1	A0	Input capacitance	22	Open	2	45	pF	41
Y2	A1	"	24	"	"	"	"	"
Y3	A2	"	26	"	"	"	"	"
Y4	A3	"	32	"	"	"	"	"
Y5	A4	"	34	"	"	"	"	"
Y6	A5	"	36	"	"	"	"	"
Y7	A6	"	37	"	"	"	"	"
Y8	A7	"	38	"	"	"	"	"
Y9	D0	"	21	"	"	17	"	"
Y10	D1	"	25	"	"	"	"	"
Y11	D2	"	31	"	"	"	"	"
Y12	D3	"	35	"	"	"	"	"
Y13	E0	"	27	"	"	45	"	"
Y14	E1	"	28	"	"	"	"	"
Y15	E2	"	29	"	"	"	"	"
Y16	$\overline{WE}$	"	39	"	"	50	"	"
Y17	Q0	Output capacitance	23	"	"	18	"	"
Y18	Q1	"	30	"	"	"	"	"
Y19	Q2	"	33	"	"	"	"	"
Y20	Q3	"	40	"	"	"	"	"

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NOTES:
1. Null out the effect of the test leads prior to connecting to the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
3. Item C01 setup:
   Frequency 1 kHz.
   V_{pk} = 0.3 V maximum.

*FIGURE 41. Power supply filter capacitance and input and output capacitance test circuit for tables XXI and XXII.

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